# Major Front-End Migration for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

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**West Coast Biological Sciences Undergraduate Research Conference**
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Loyola Marymount University
Frank R. Seaver College of Science and Engineering

## Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein-Protein Interactions (PPI).
- GRNsight is an integral part of a data analysis workflow for biologists who model GRNs.
- GRNsight's interface was originally built using older web tools, leading to repeated code and difficult user interface updates.
- I have migrated GRNsight's user interface to React Vite and Grommet UI which are modern, scalable, and industry-standard frameworks.
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## Gene expression is the flow of information in a cell from DNA to RNA to protein

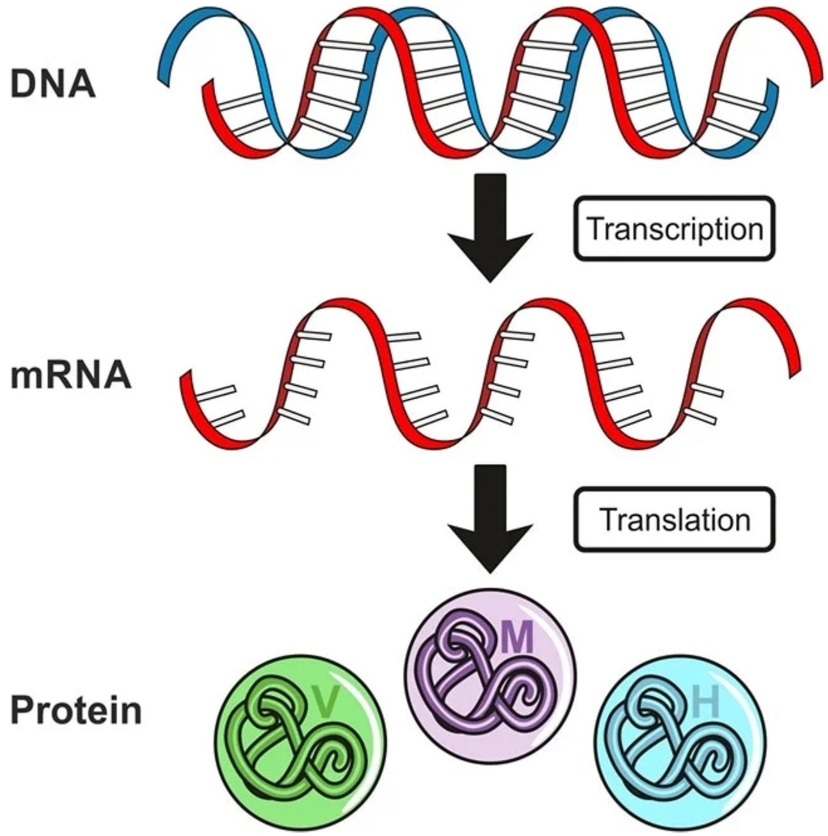
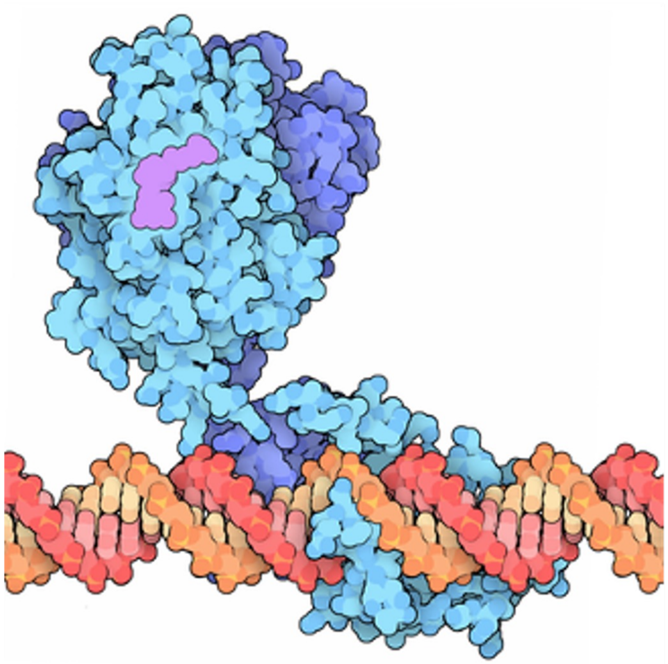



Image from Dr. Surat P, Ph.D. on AZO Life Sciences

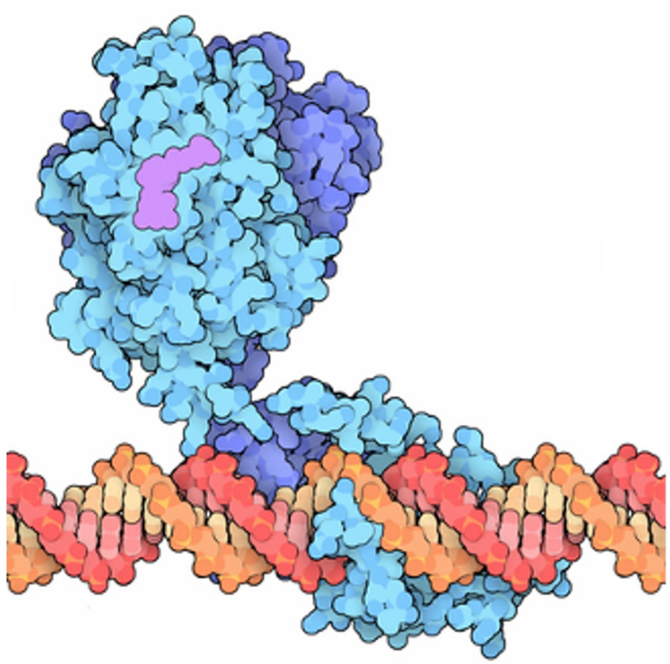
## Regulatory Transcription Factors bind to DNA and turn on (activate) or turn off (repress) gene expression



“PDB101: Molecule of the Month: Vitamin D receptor,” RCSB: PDB-101. https://pdb101.rcsb.org/motm/155

- In yeast ~250 regulatory transcription factors regulate ~6000 genes.
- In humans, ~500 regulatory transcription factors regulate ~20,000 genes

## A Gene Regulatory Network (GRN) consists of genes, transcription factors and the regulatory connections between them which govern the level of expression of mRNA and protein from genes



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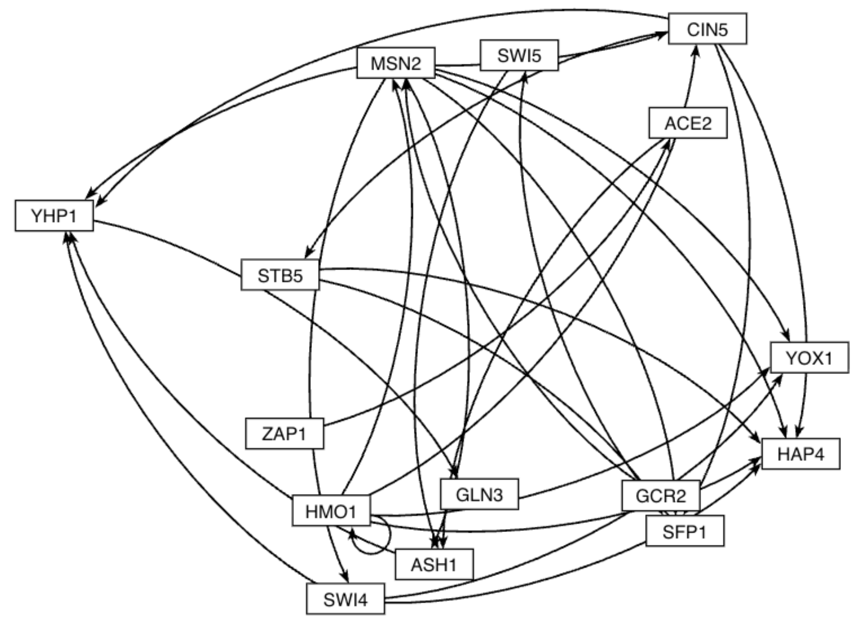
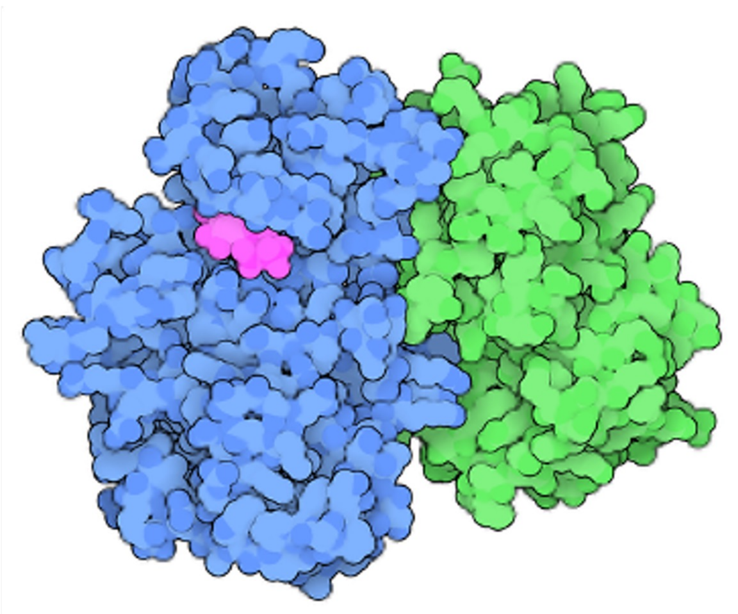
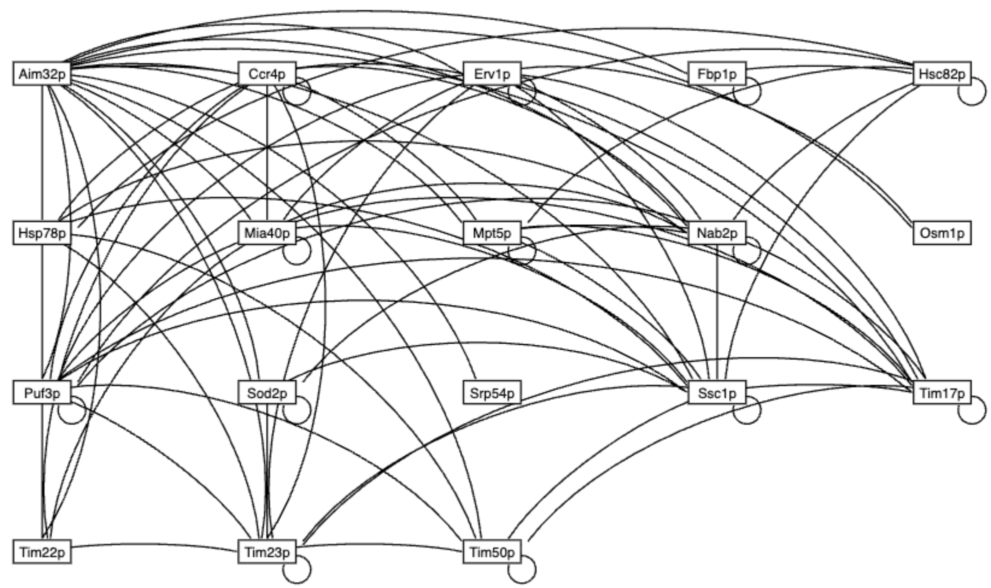



Yeast GRN that controls the cold shock response
- Node = transcription factor / gene
- Arrow edge = regulatory relationship

## Protein-Protein Interactions refer to physical interactions of proteins binding to each other - the basis of nearly all cell functions



“PDB101: Molecule of the Month: Cyclin and cyclin-dependent kinase,” RCSB: PDB-101. https://pdb101.rcsb.org/motm/236

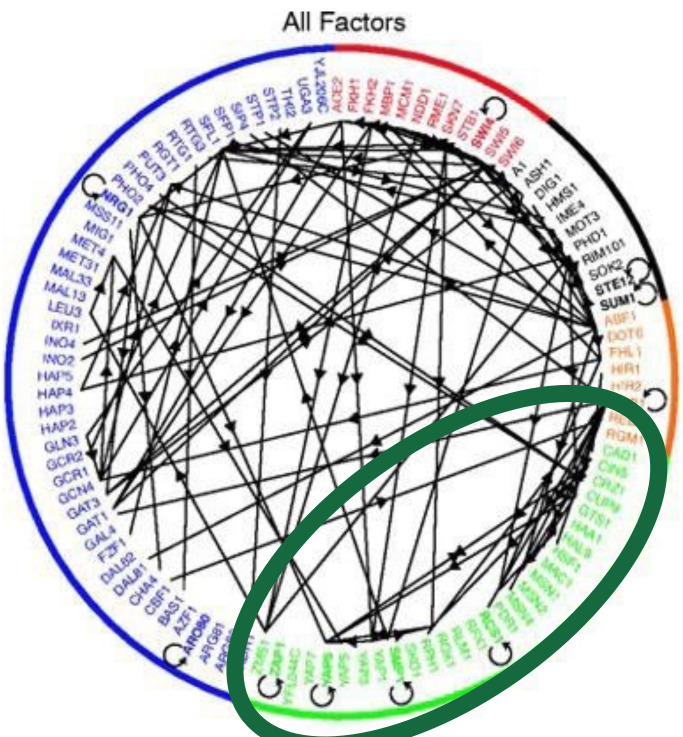



PPI network for yeast mitochondrial protein Aim32p

- Node = proteins
- Edge = binding relationship

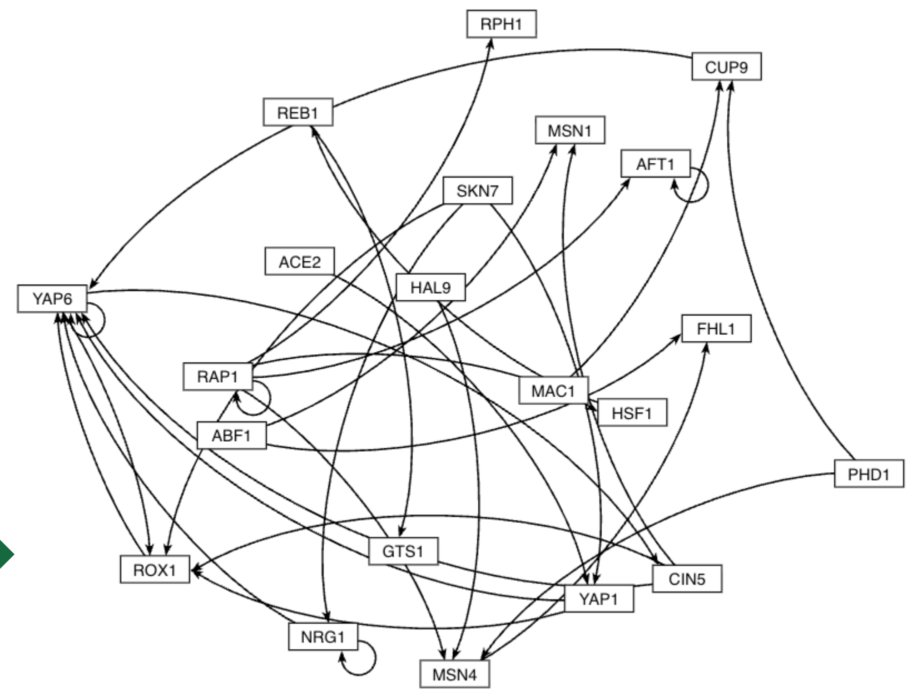
# Large gene regulatory networks are difficult to visualize, and GRNsight makes it easier to visualize networks of 35 nodes or fewer

**GRN of 106 transcription factors in yeast**




**Lee *et al.* (2002)**

**GRN subnetwork of 21 transcription factors that regulate the environmental stress response in yeast**




**GRNsight**

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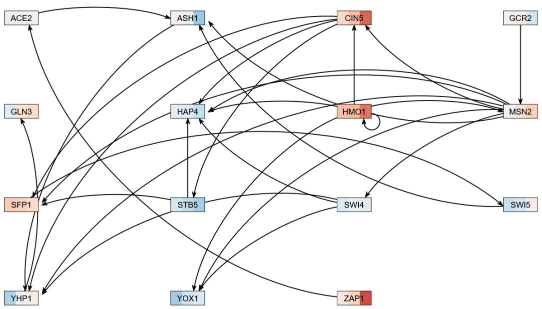
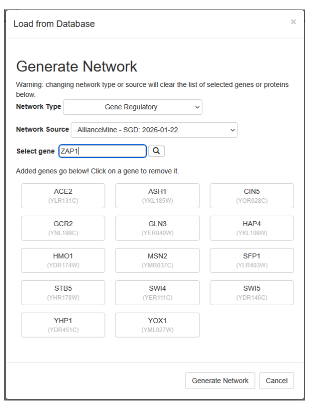
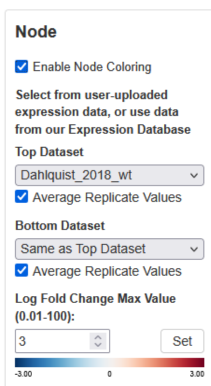



Select network and expression data from GRNsight database



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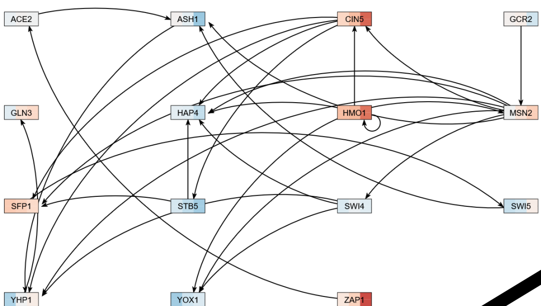


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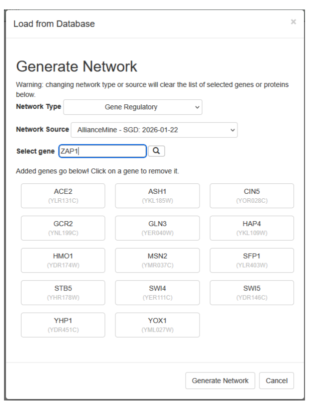
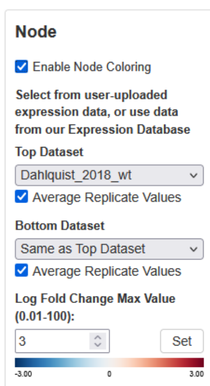
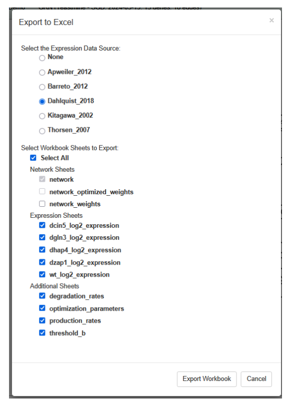
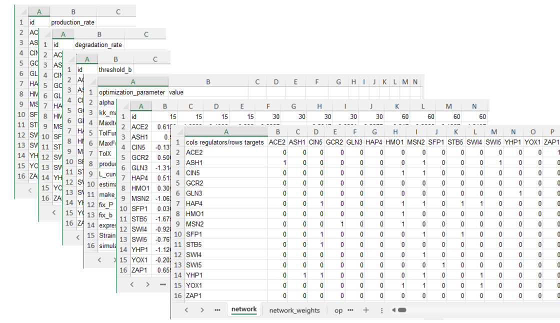



Export to Excel



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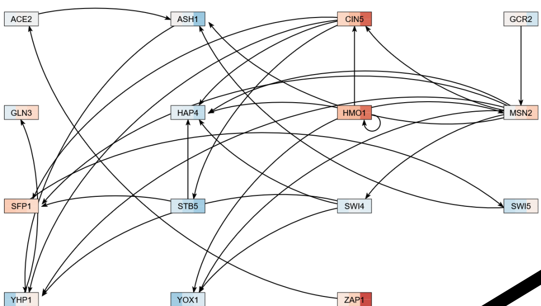


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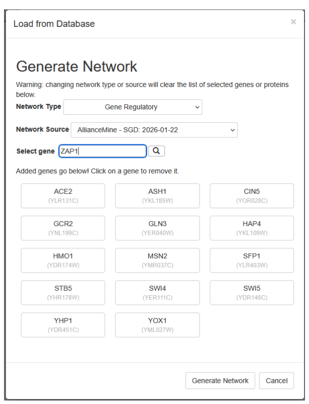
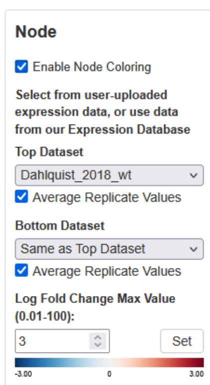
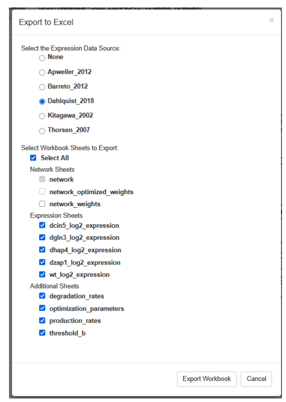
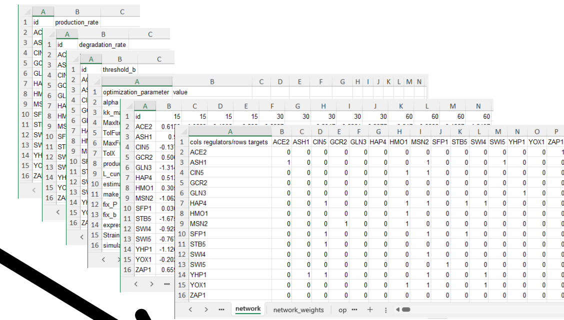


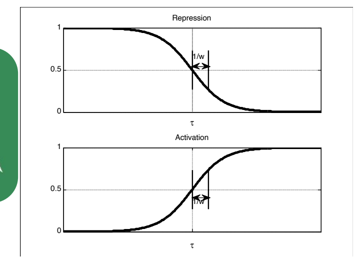




**Export to Excel**

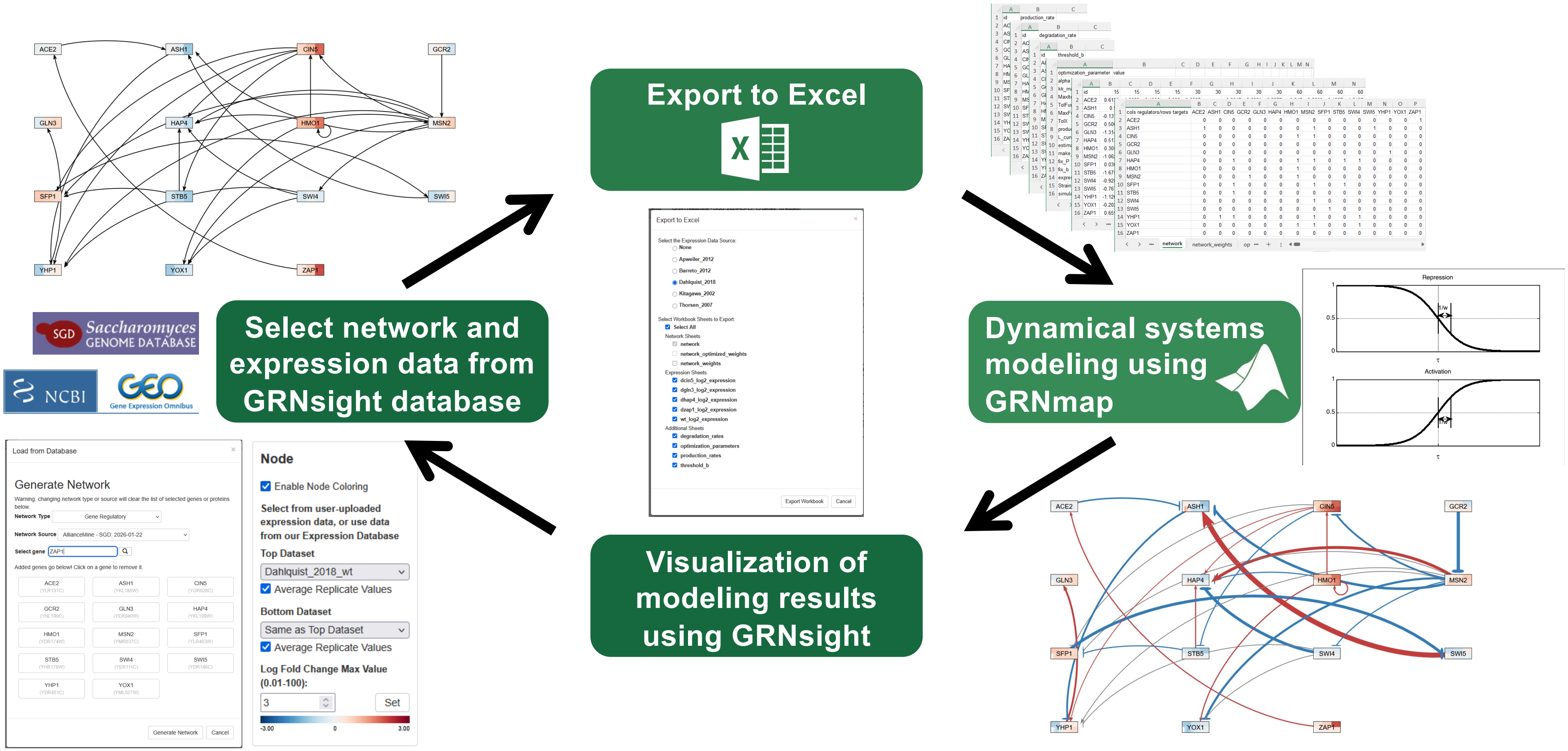


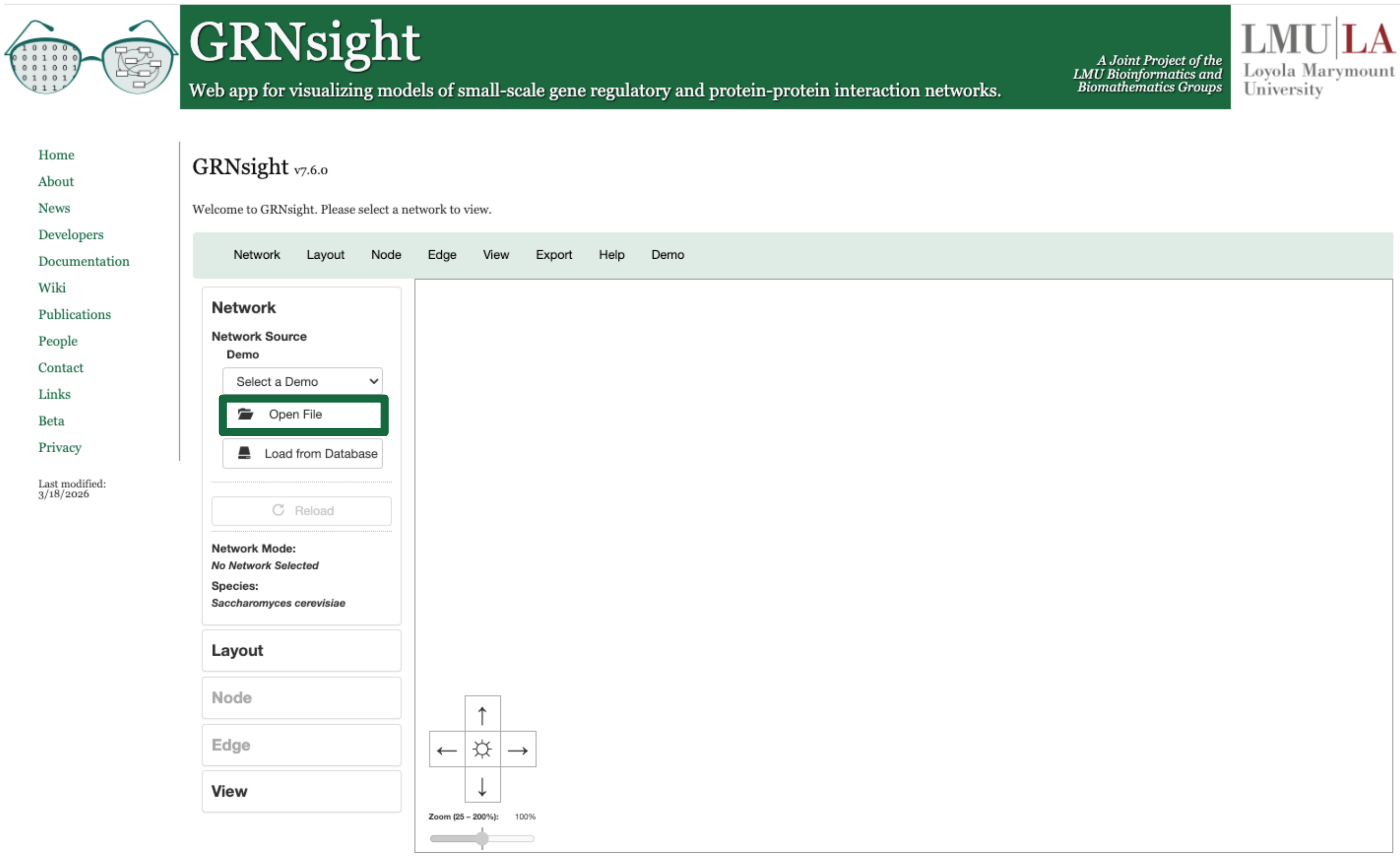
**Dynamical systems modeling using GRNmap**



# GRNsight is an integral part of a data analysis workflow for biologists who model GRNs

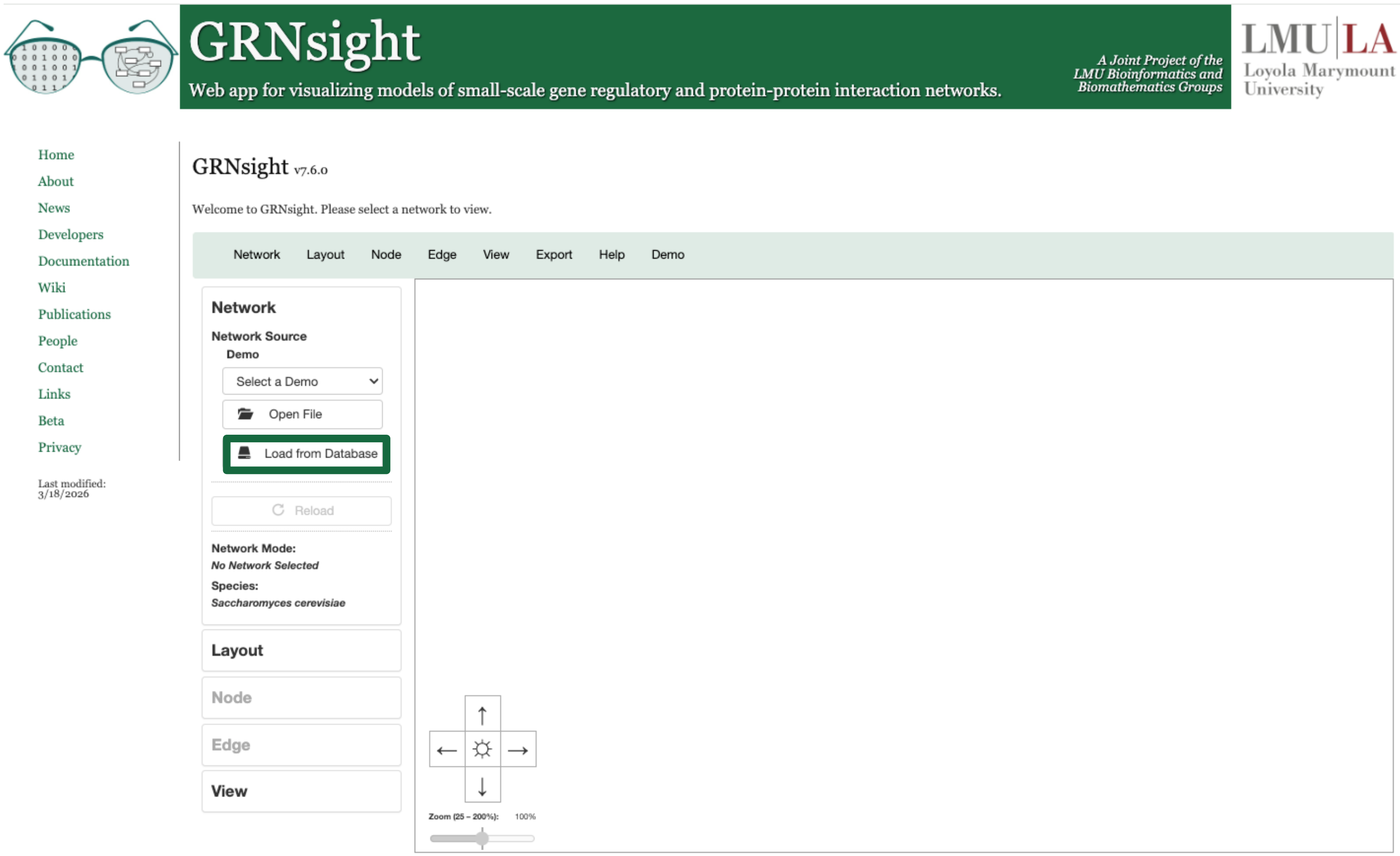


# Users can import networks via the file upload functionality in a supported format (SIF, GraphML, or Excel)

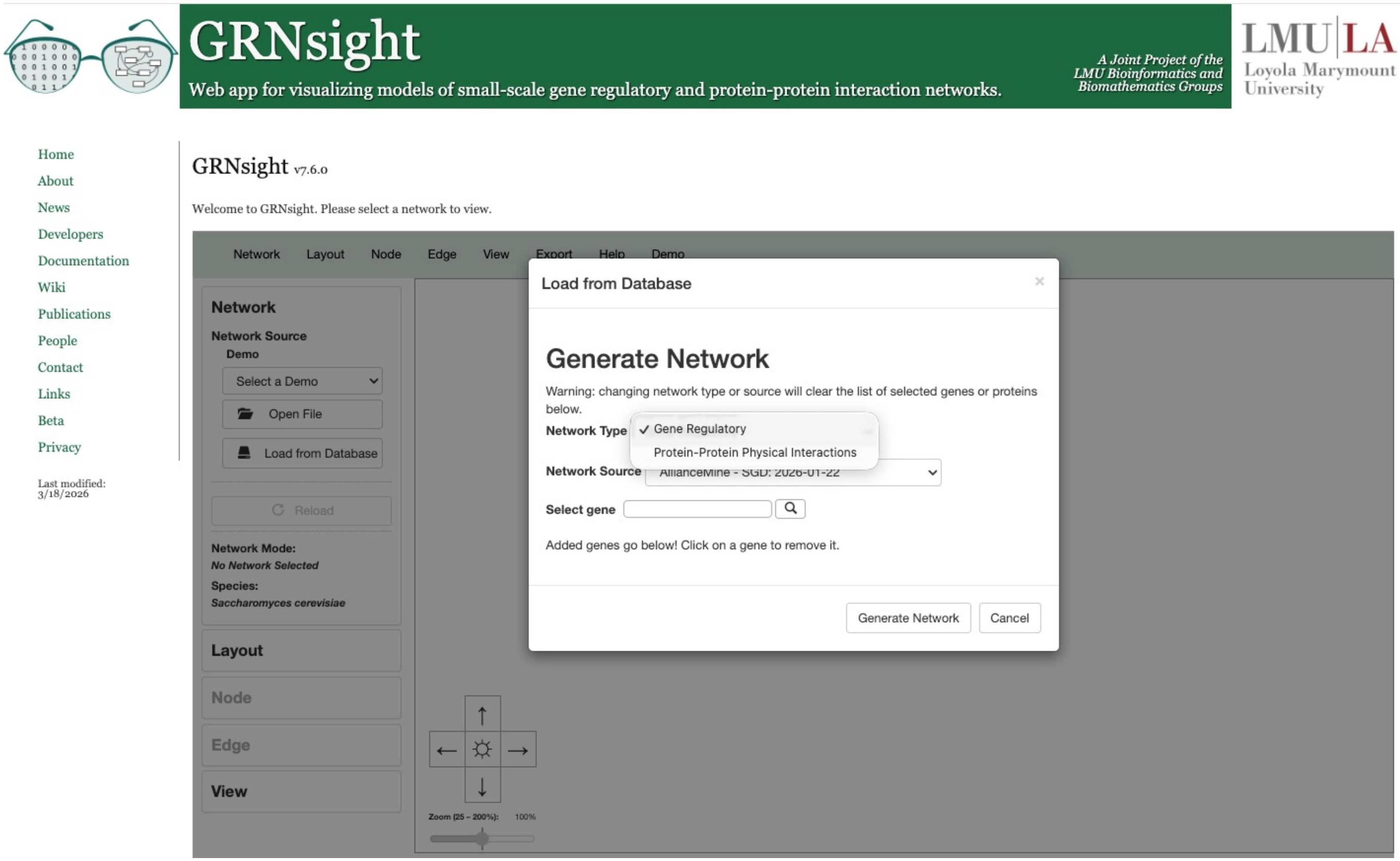



Dahlquist et al. (2016) *PeerJ Computer Science*; Version 1 available March 20, 2014

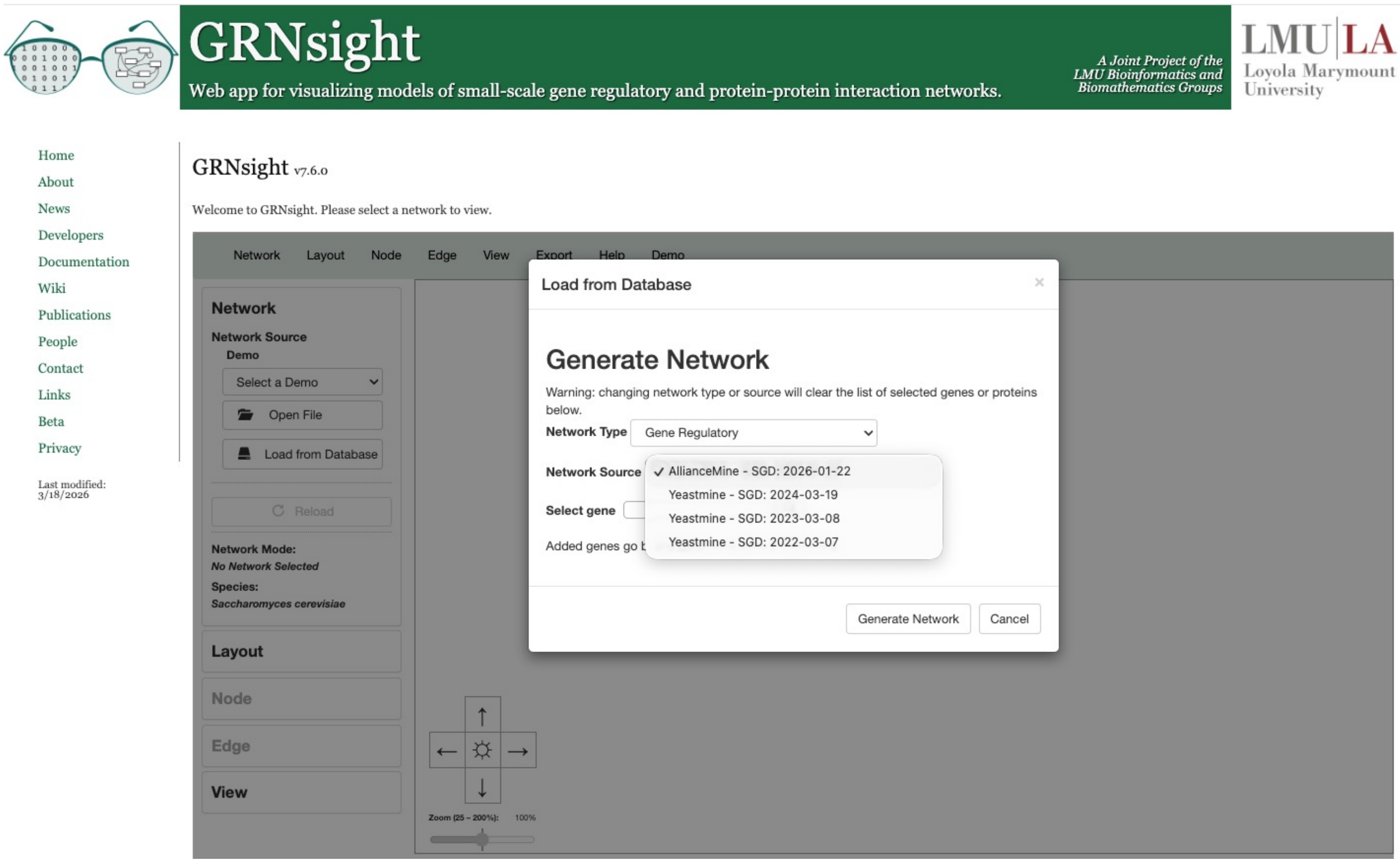
# Users can load networks from our database, which contains regulation data from the *Saccharomyces* Genome Database (SGD)



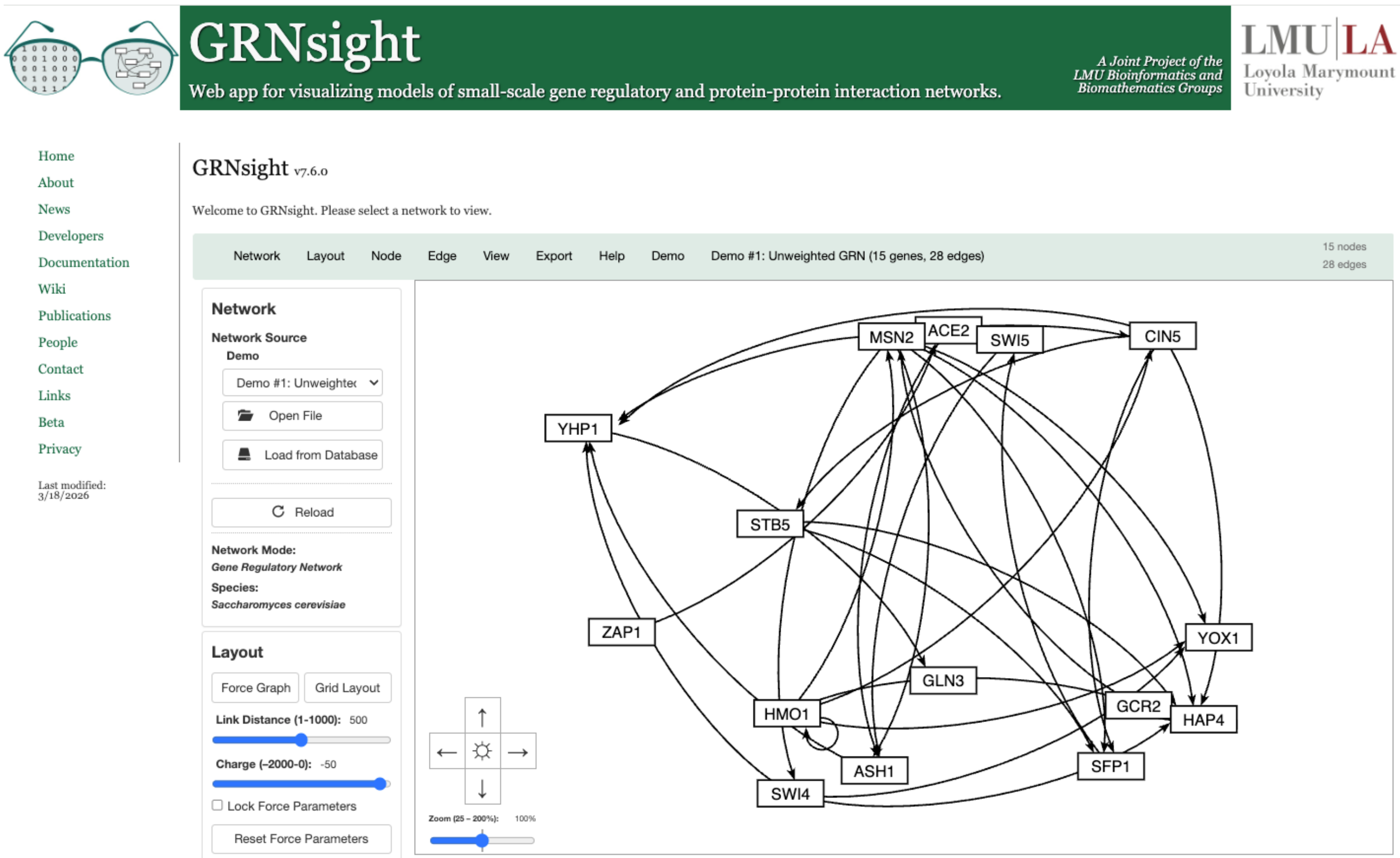
# Users can load different network types, either GRN or PPI



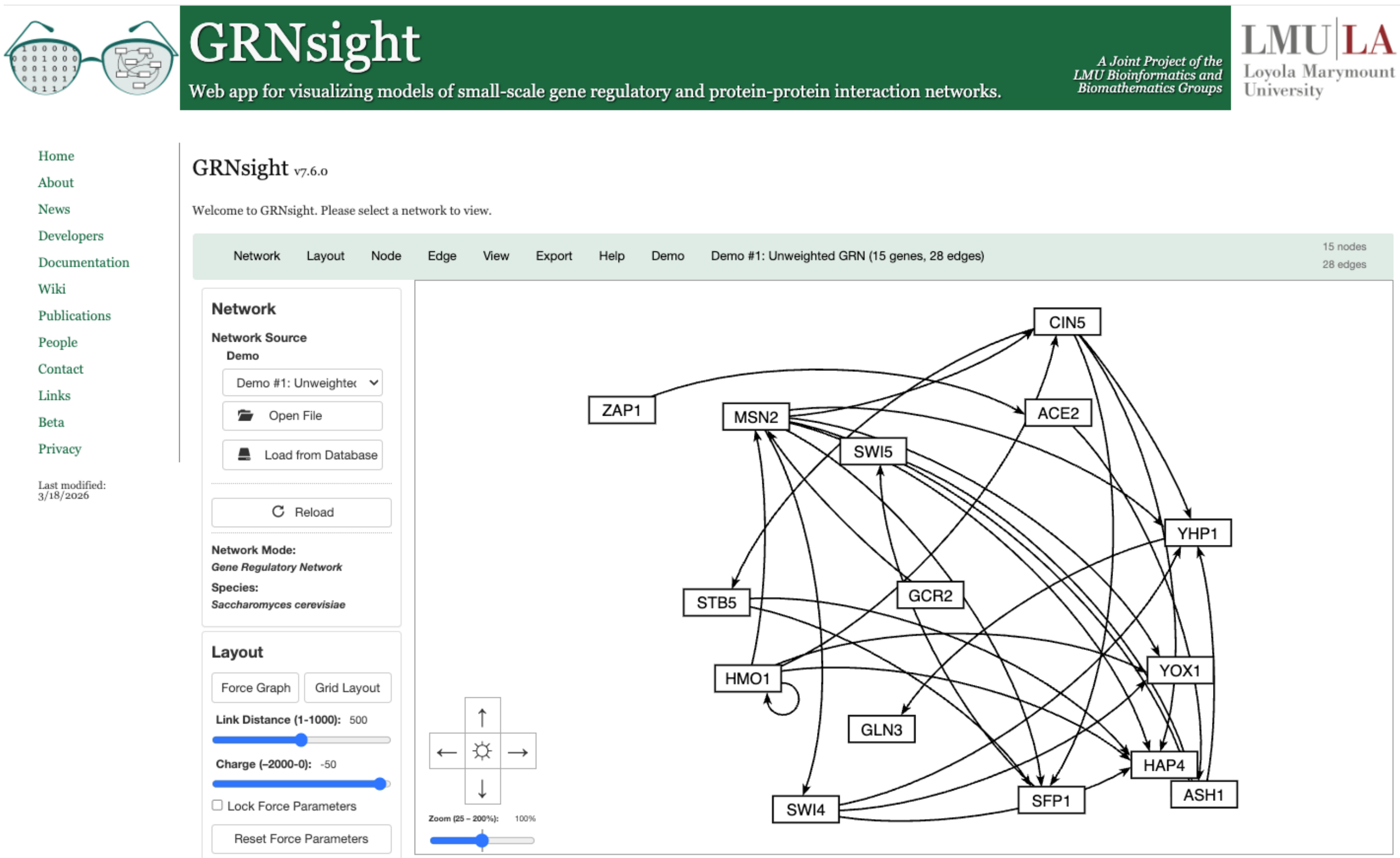
# Users can choose different network sources since GRNsight refreshes with new data each year



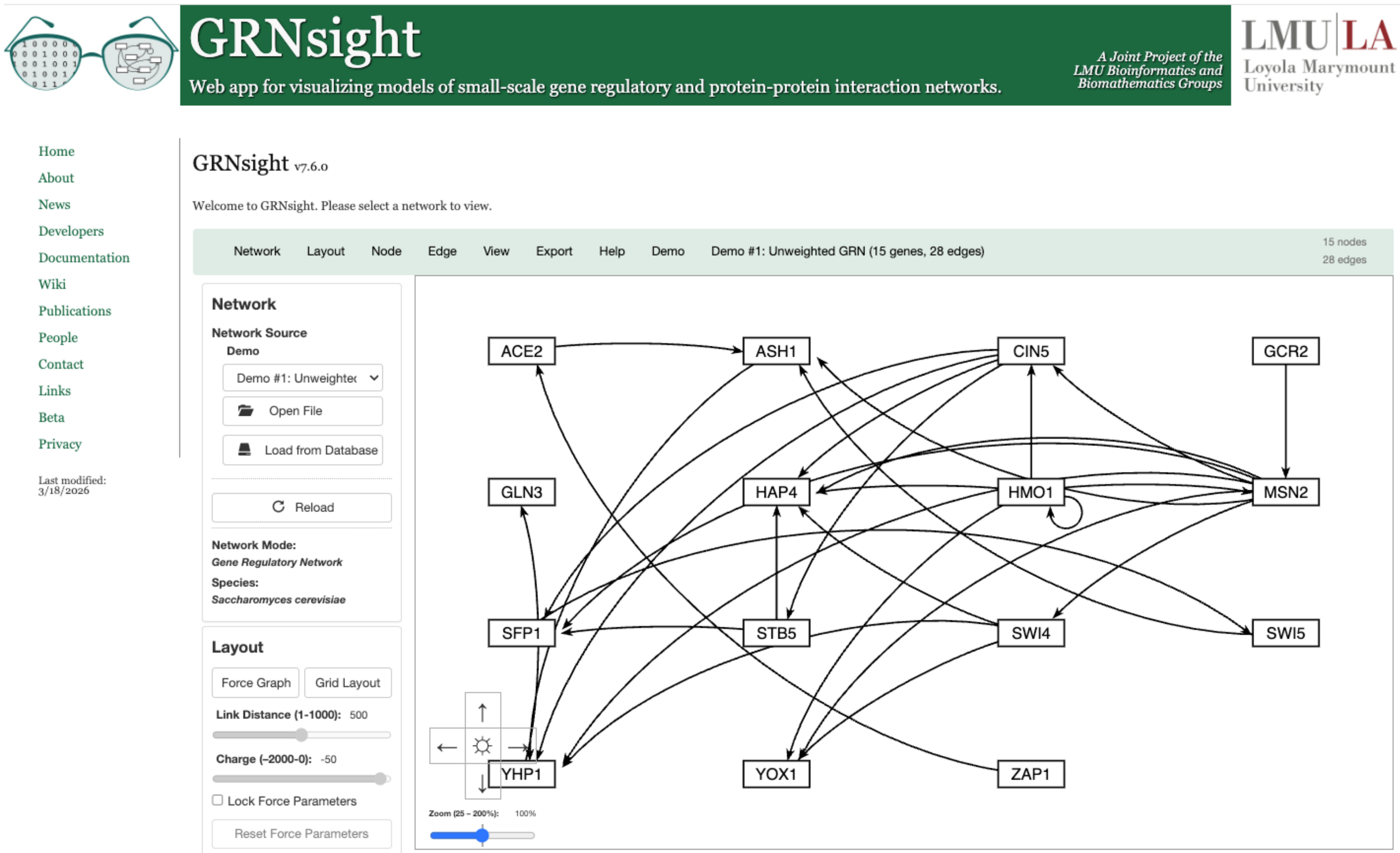
# GRNsight rapidly generates GRN graphs using our customizations to the open source D3 library



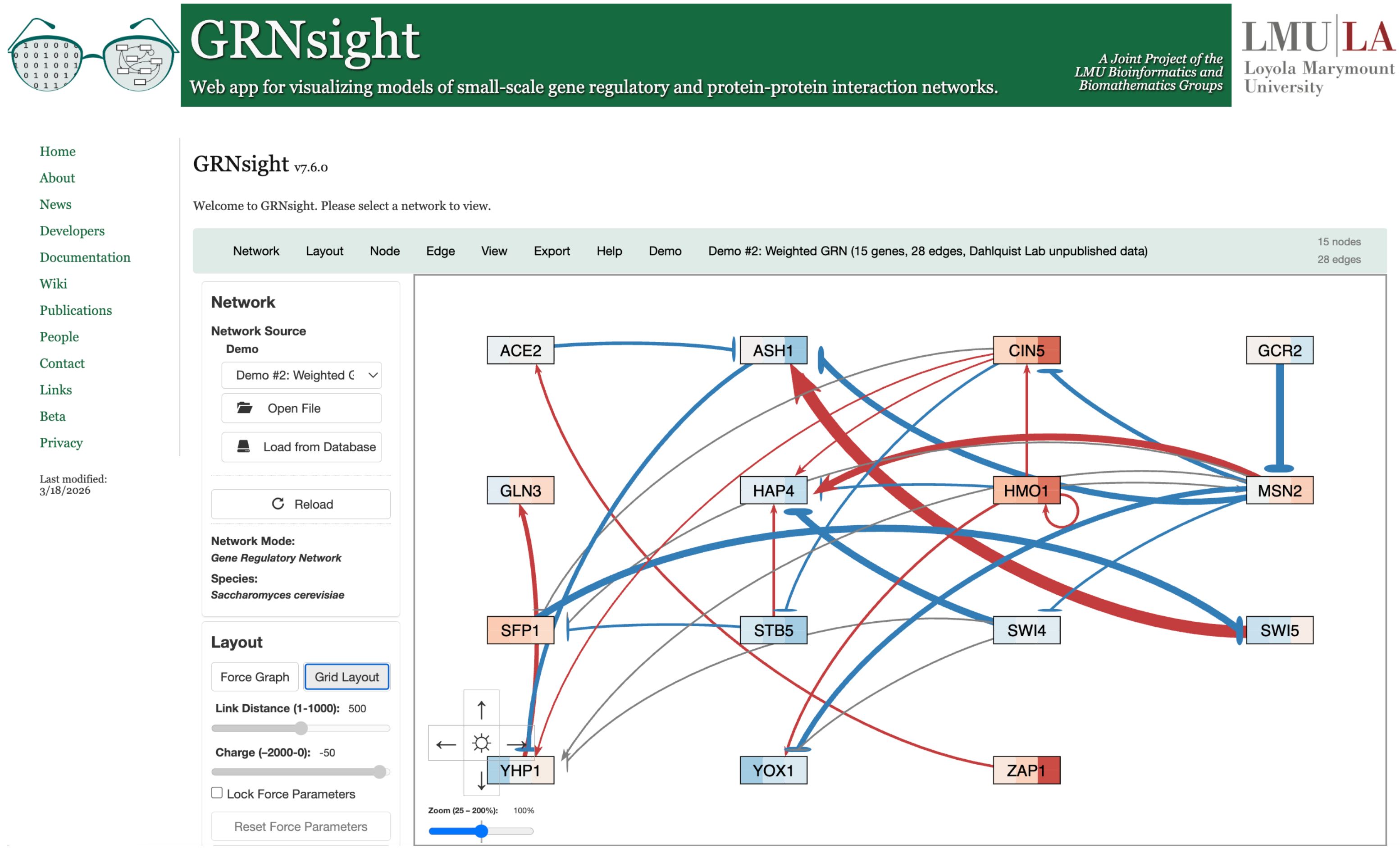
# Besides the Force Graph Layout, Users Can Rearrange the Nodes by Hand



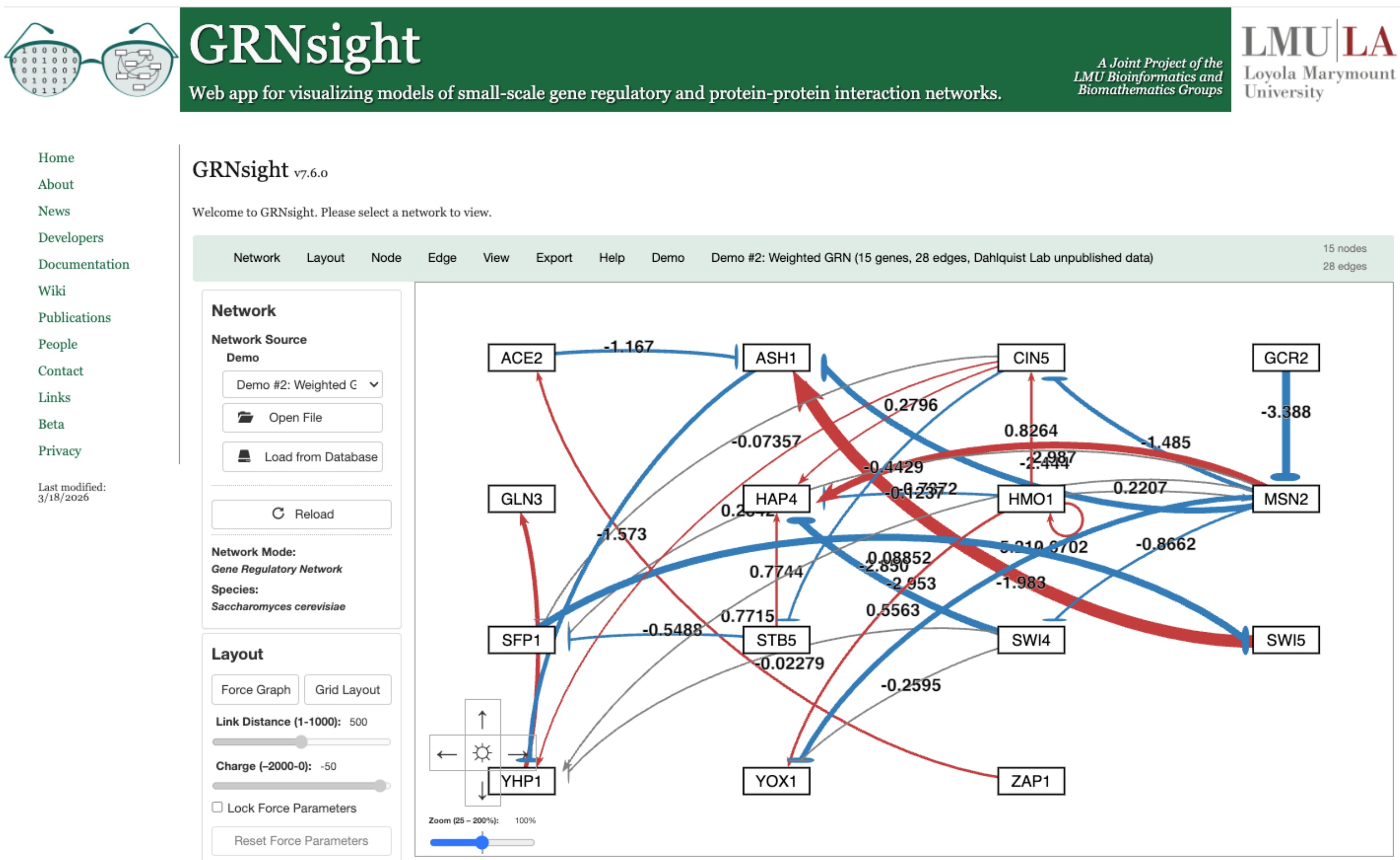
# Or Choose the Convenient Block Layout Option



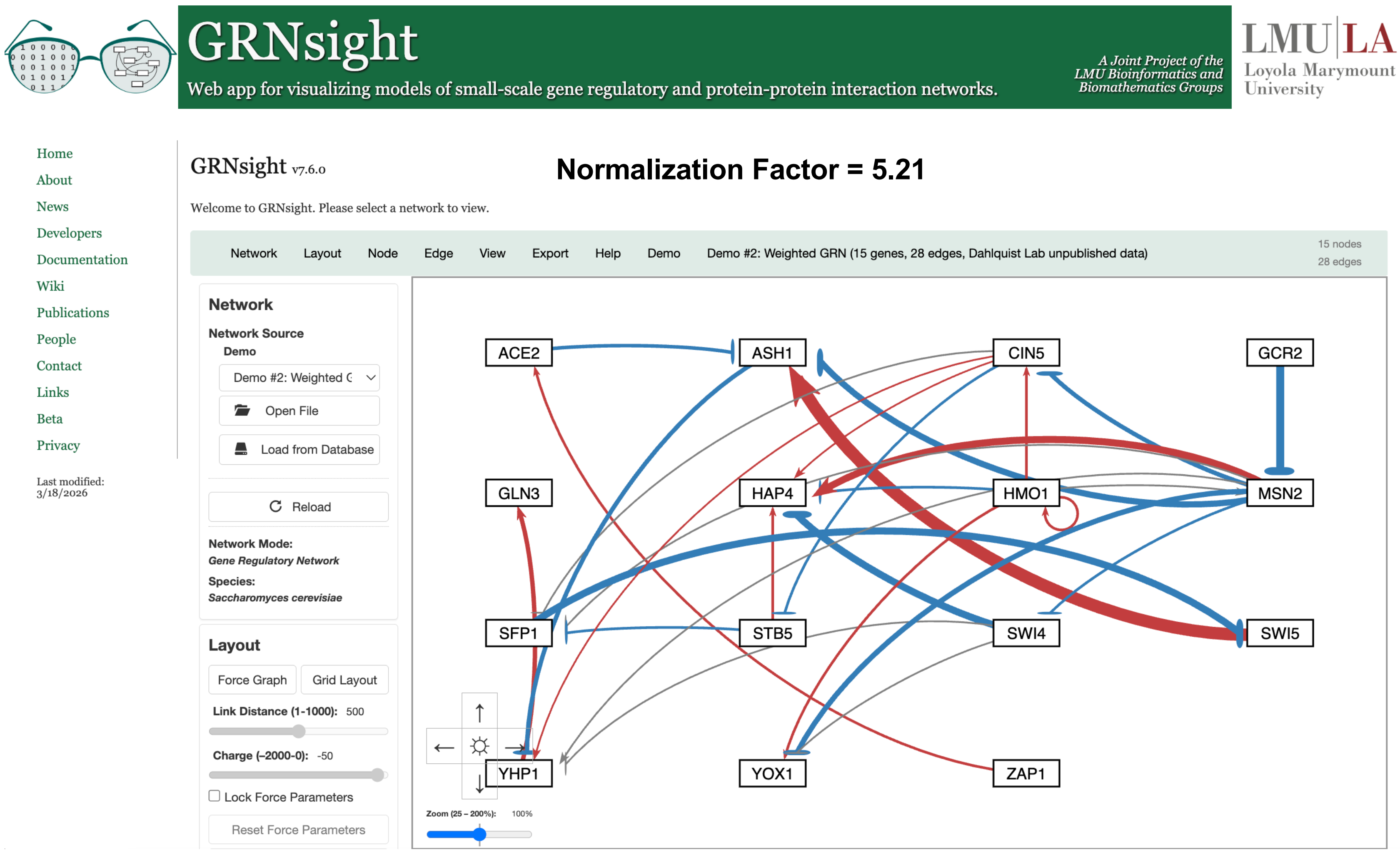
# When Model Results are Loaded, Edge Weight Values Are Indicated by Color, Thickness, and End Marker



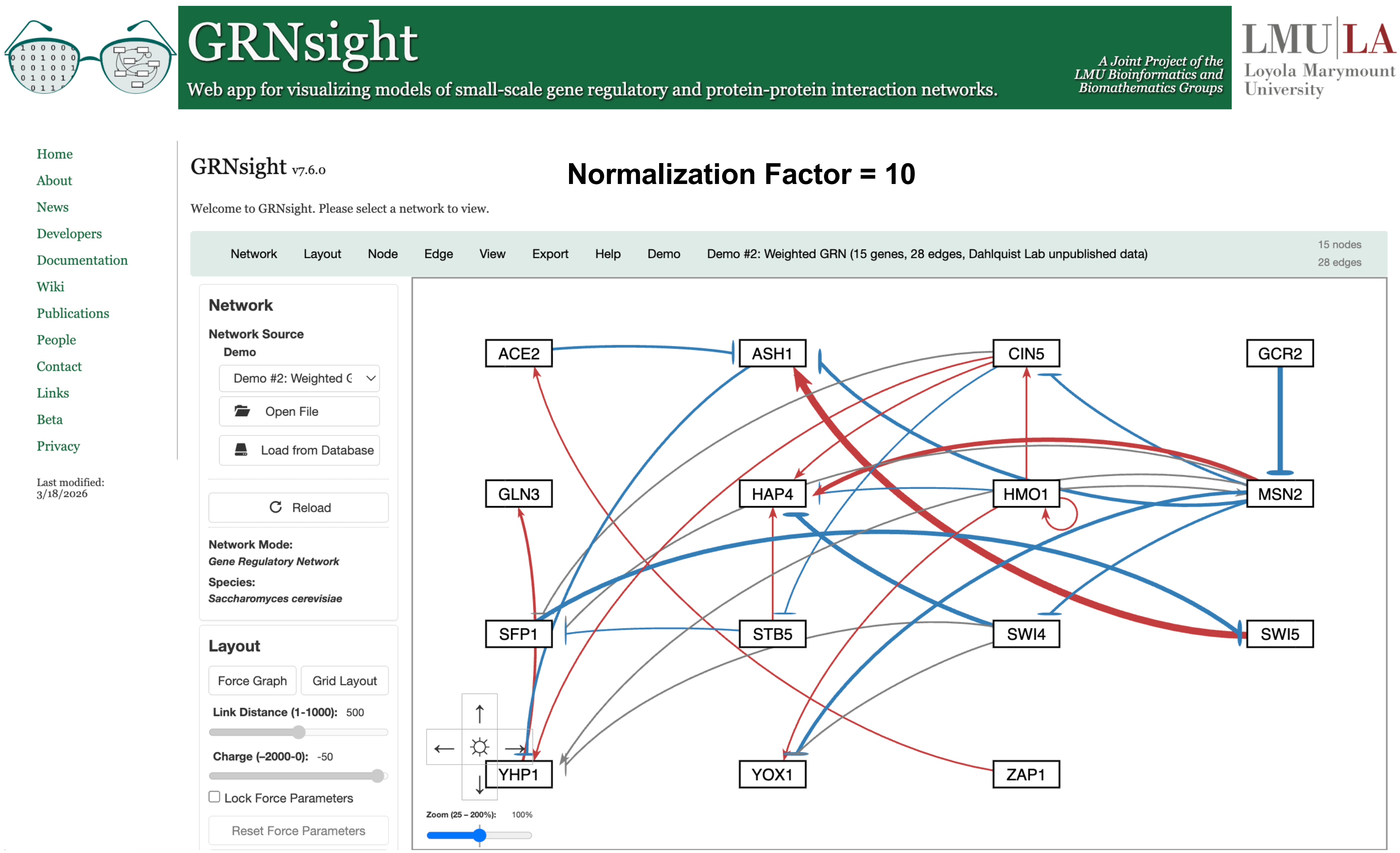
# Numerical Weight Values Can Be Displayed on the Graph



# Edge Thicknesses are Normalized to the Largest Magnitude Weight Value



# And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



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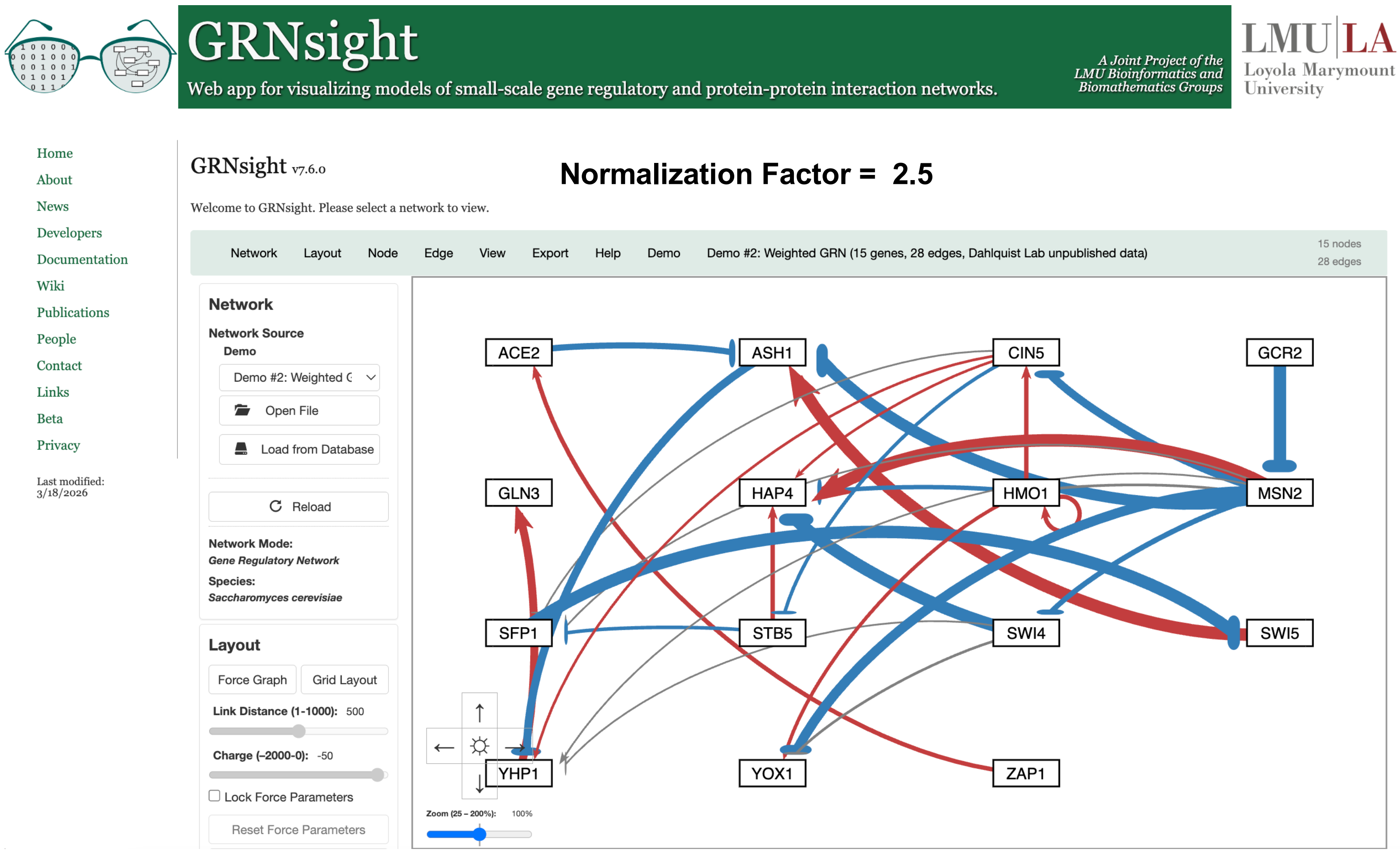
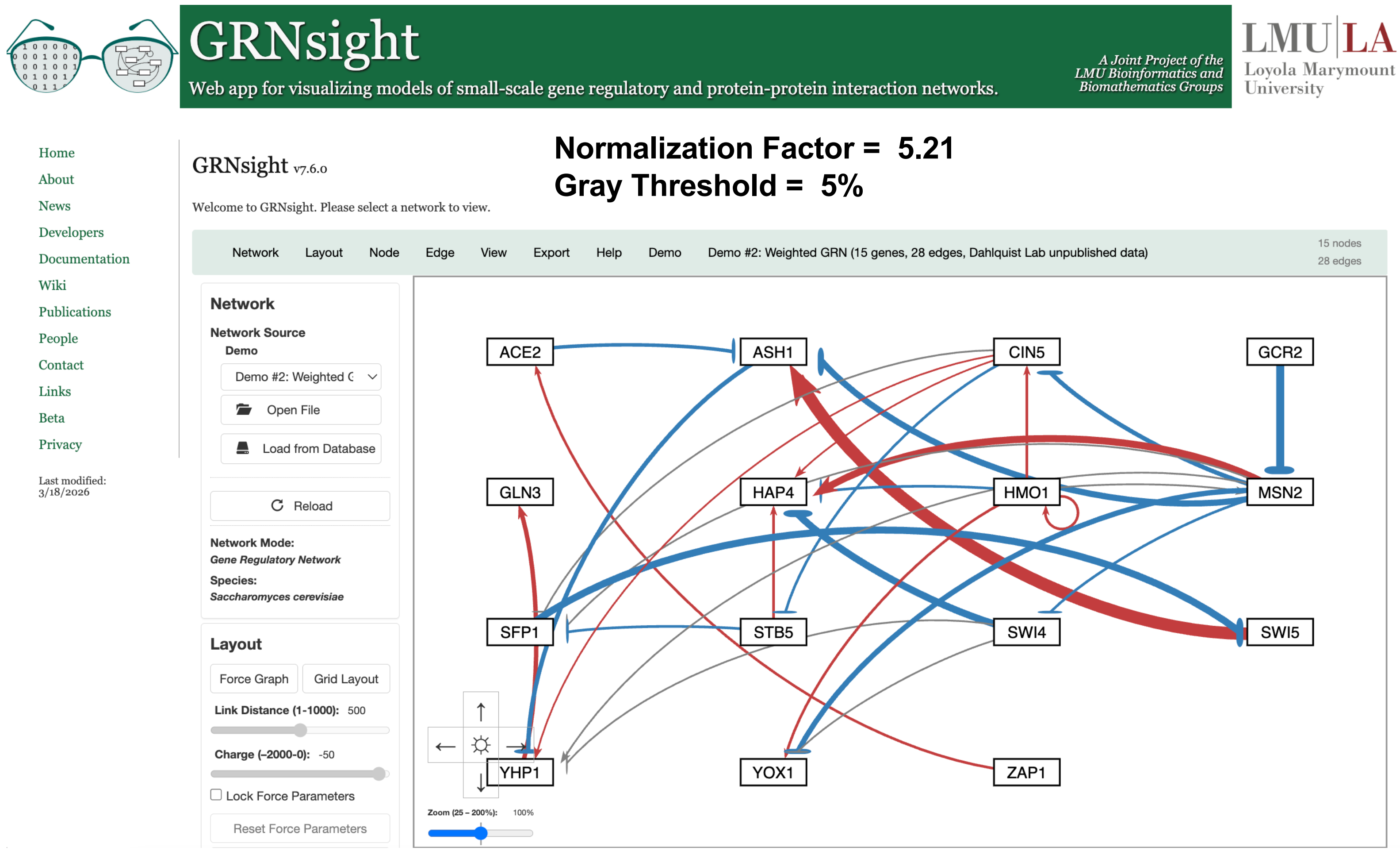
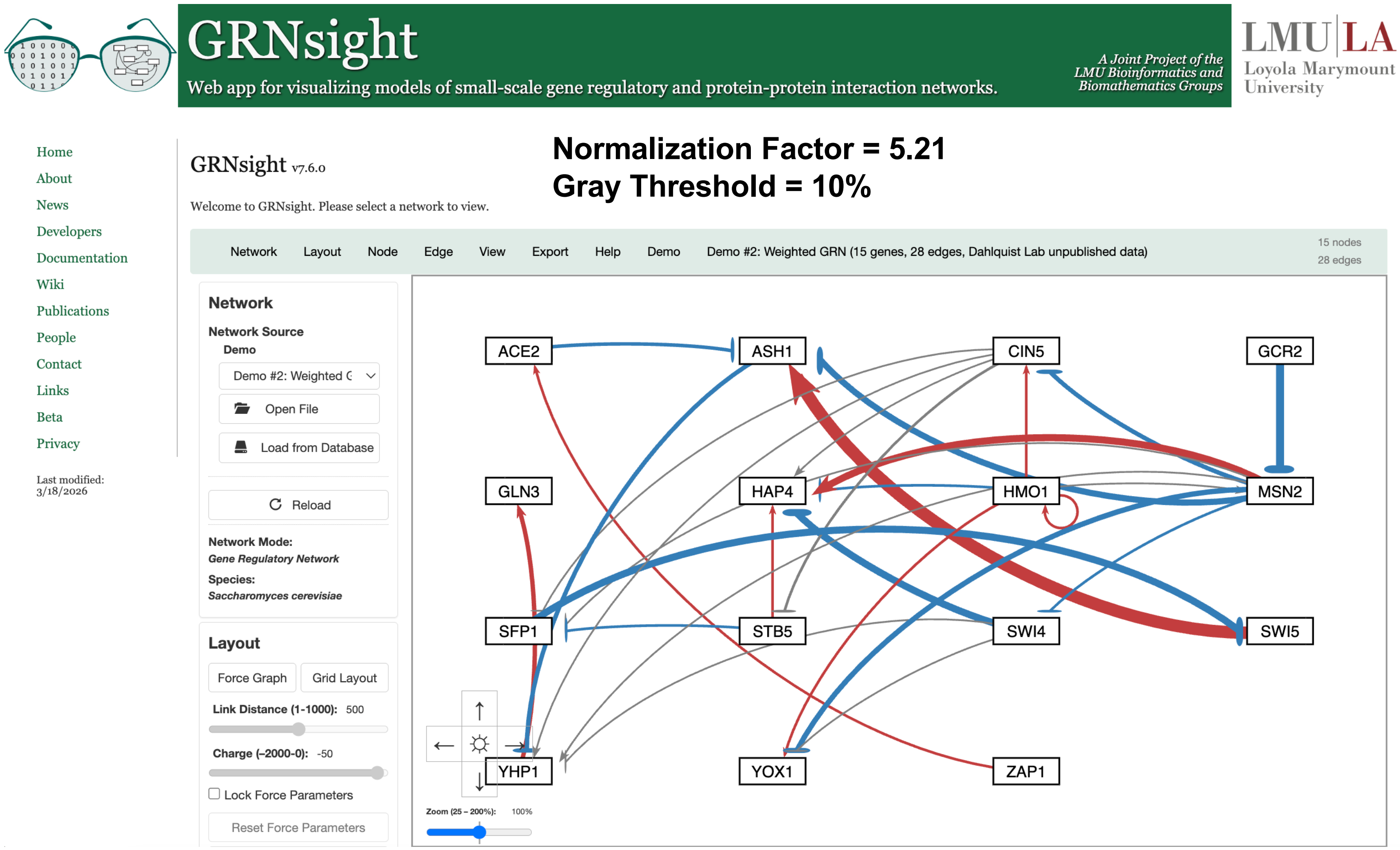


# Edges with Normalized Weight Values within 5% of 0
# Are Displayed as Gray to Minimize Their Visual Strength



# Users Can Change the Gray Threshold with a Slider

**Normalization Factor = 5.21**
**Gray Threshold = 10%**



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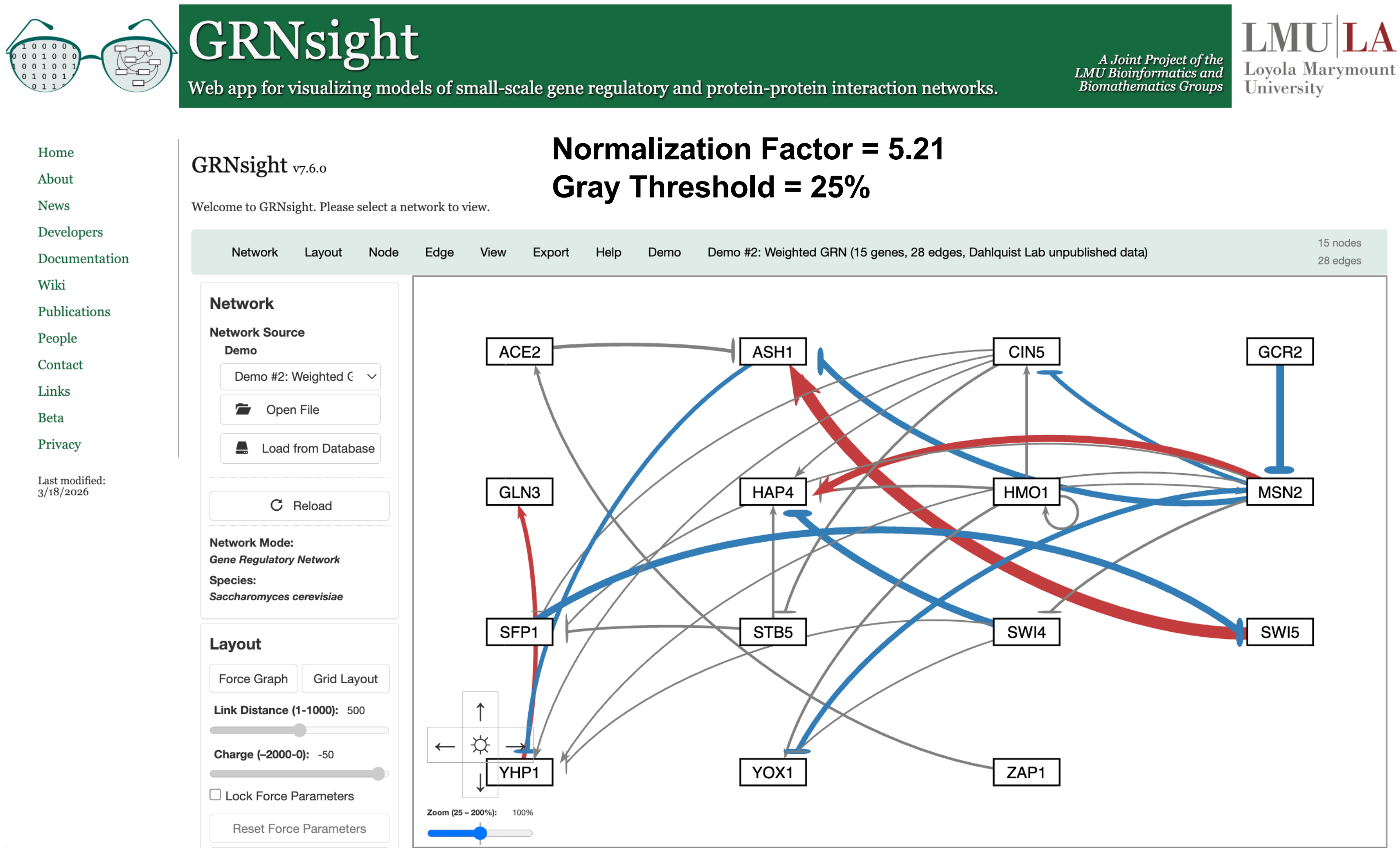

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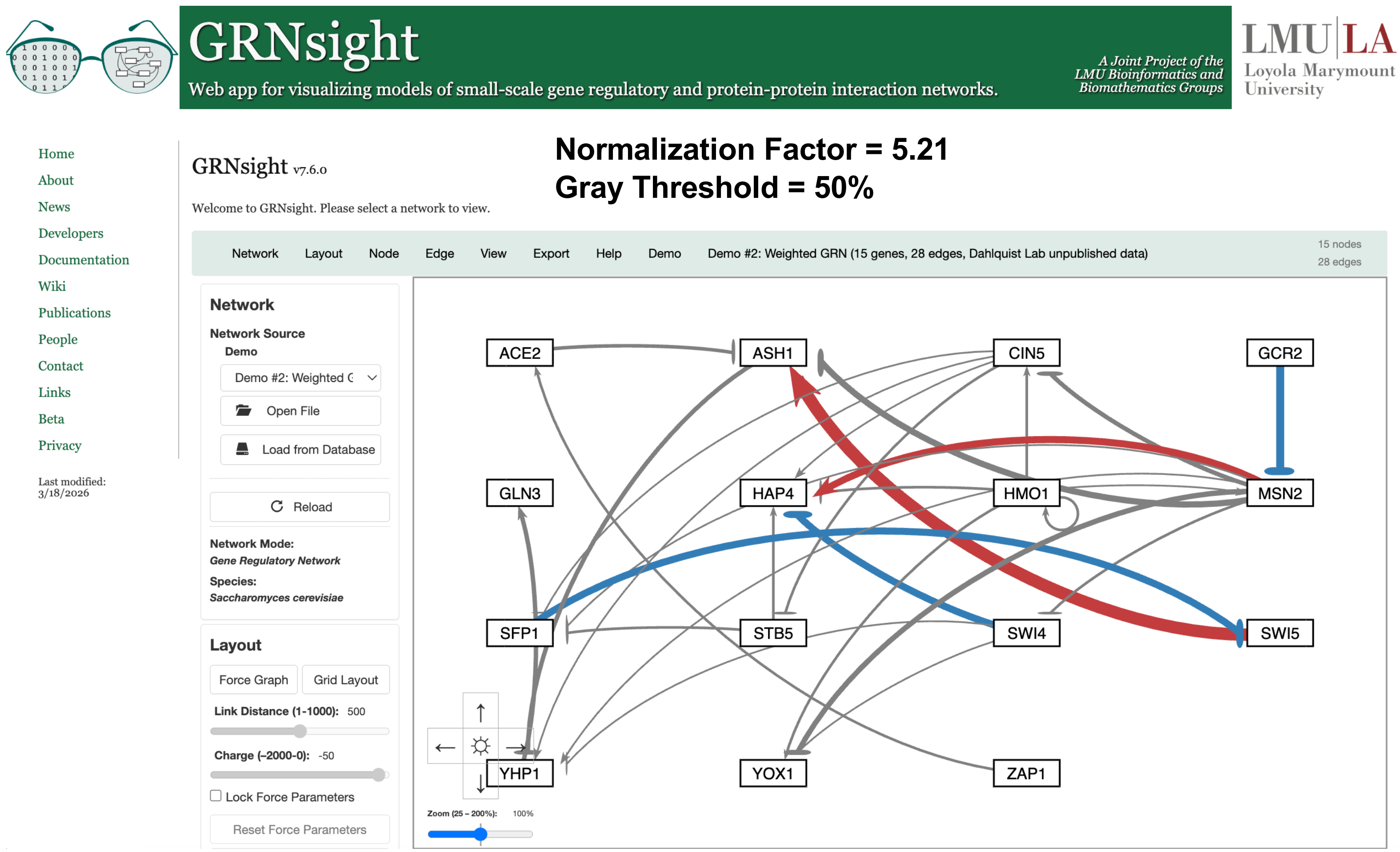
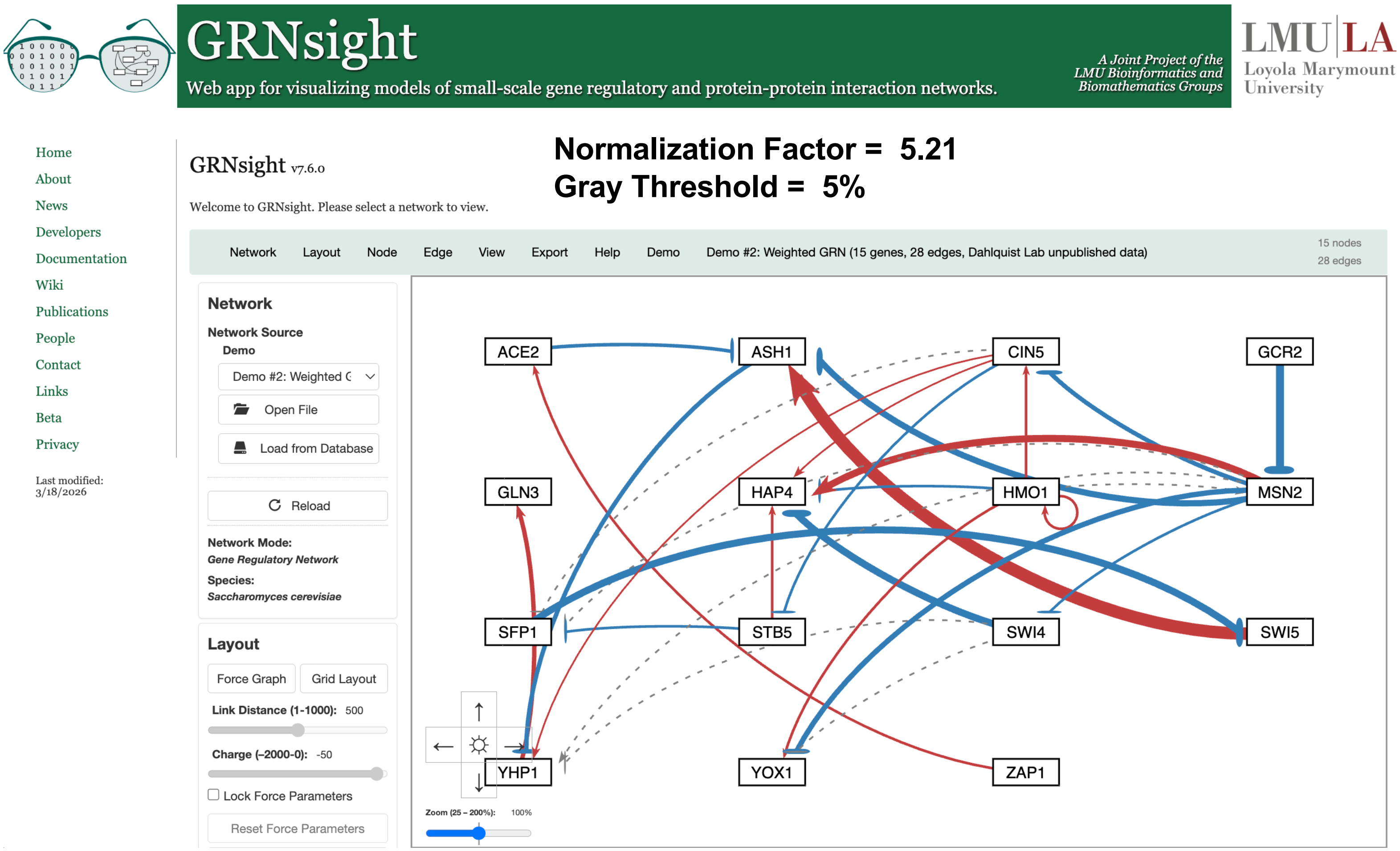
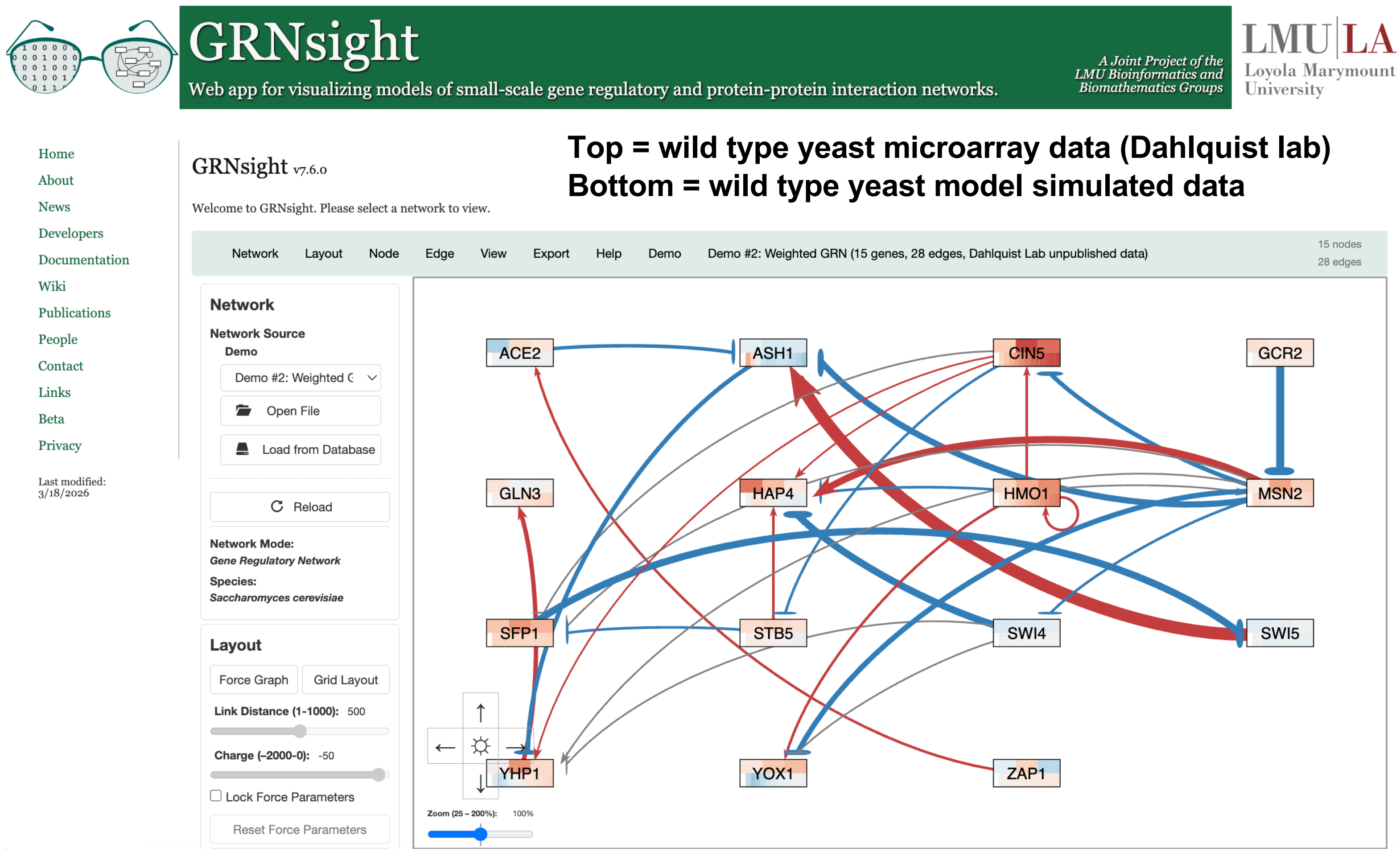


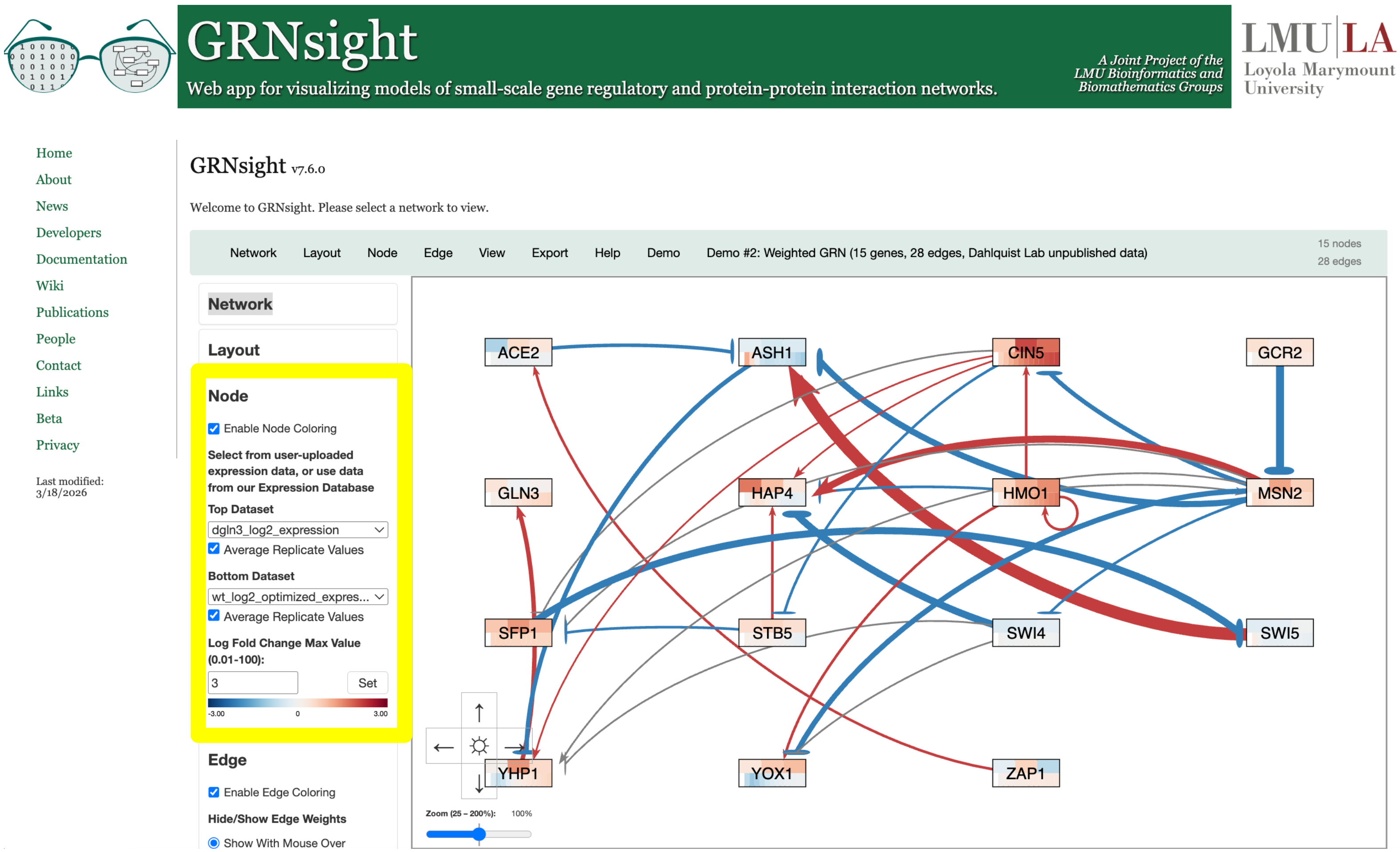
# Gray Edges Can Also Be Shown as Dashed Lines to Facilitate Visualization by Color Blind Users



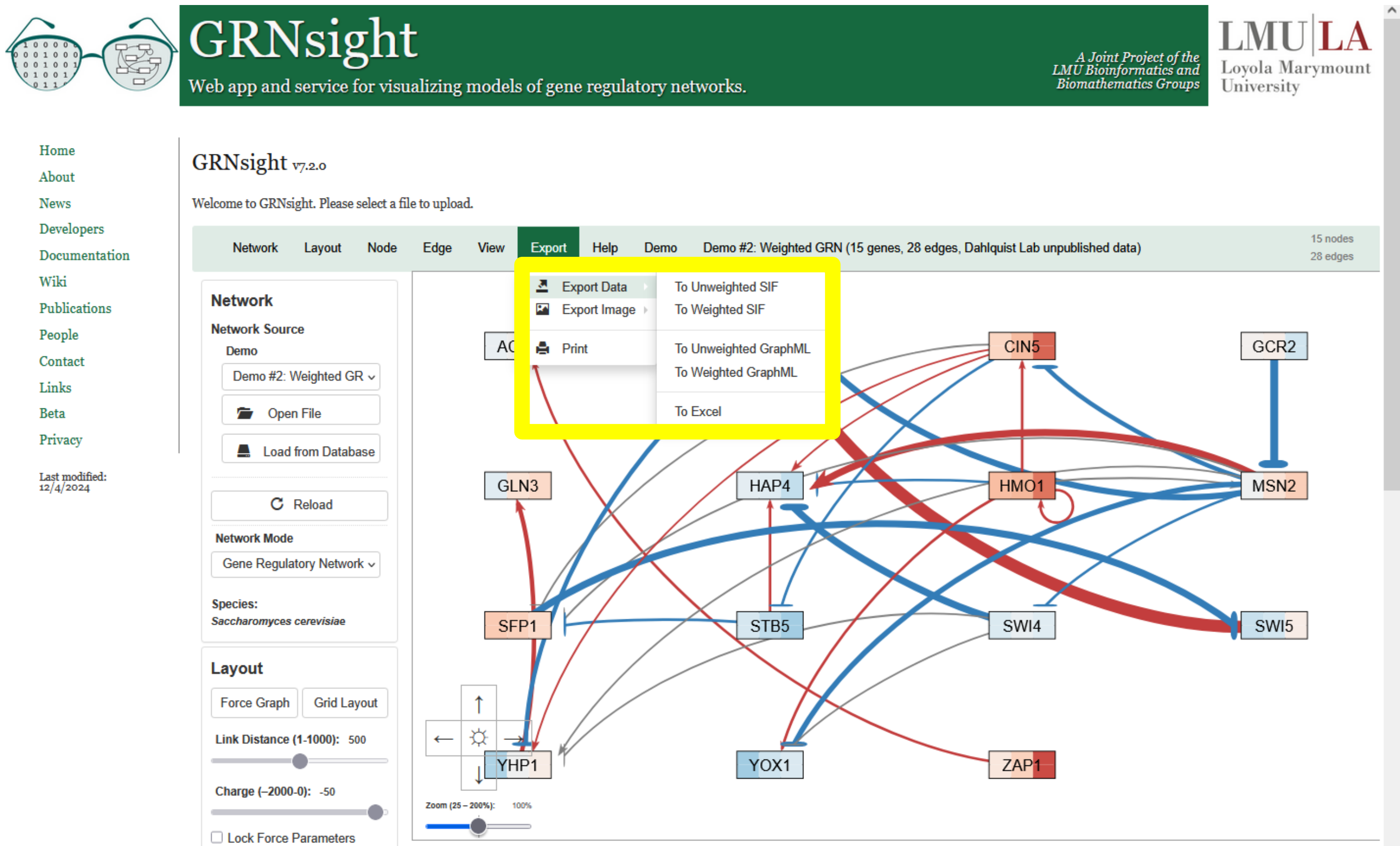
# Nodes Can Display the Measured and/or Simulated Expression Data Uploaded by the User as Individual Heat Maps



# Additional datasets from a backend PostgreSQL expression database can be used to color nodes



# Network and expression data can be exported to a GRNmap-compatible Excel Workbook for modeling



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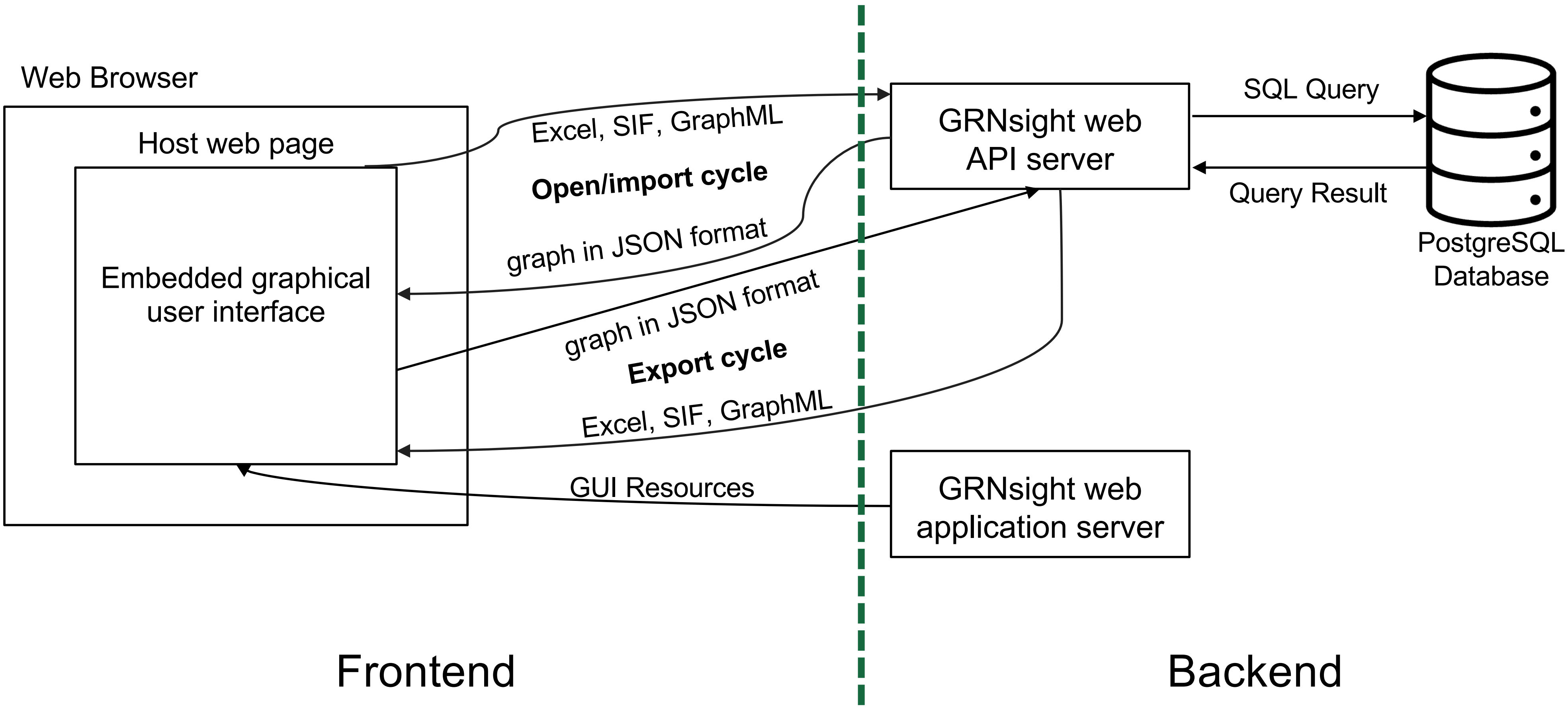
## GRNsight separates what users see (frontend), from how the data is handled (backend)



## React migration resulted in tech stack improvements for developer experience and shorter build time

| Feature | Classic | React / New Updates |
|---|---|---|
| Frontend | Bootstrap, pug, jQuery, D3 | grommet, React, D3 |
| Backend | node JS, PostgreSQL, express, JS | node JS, PostgreSQL, express, JS |
| Build Tool | webpack | Vite |
| Testing | MOCHA, chai | Vitest |
| State Management | Manual tracking of user interaction and input, and interface updates are tedious. | Automatic re-rendering of app based on user interaction and input, simplifying code. |
| Bug Fixes | Repetitive and complicated code makes bugs more likely and harder to find. | Modular code allows for easier discovery of bugs and fewer bugs created. |

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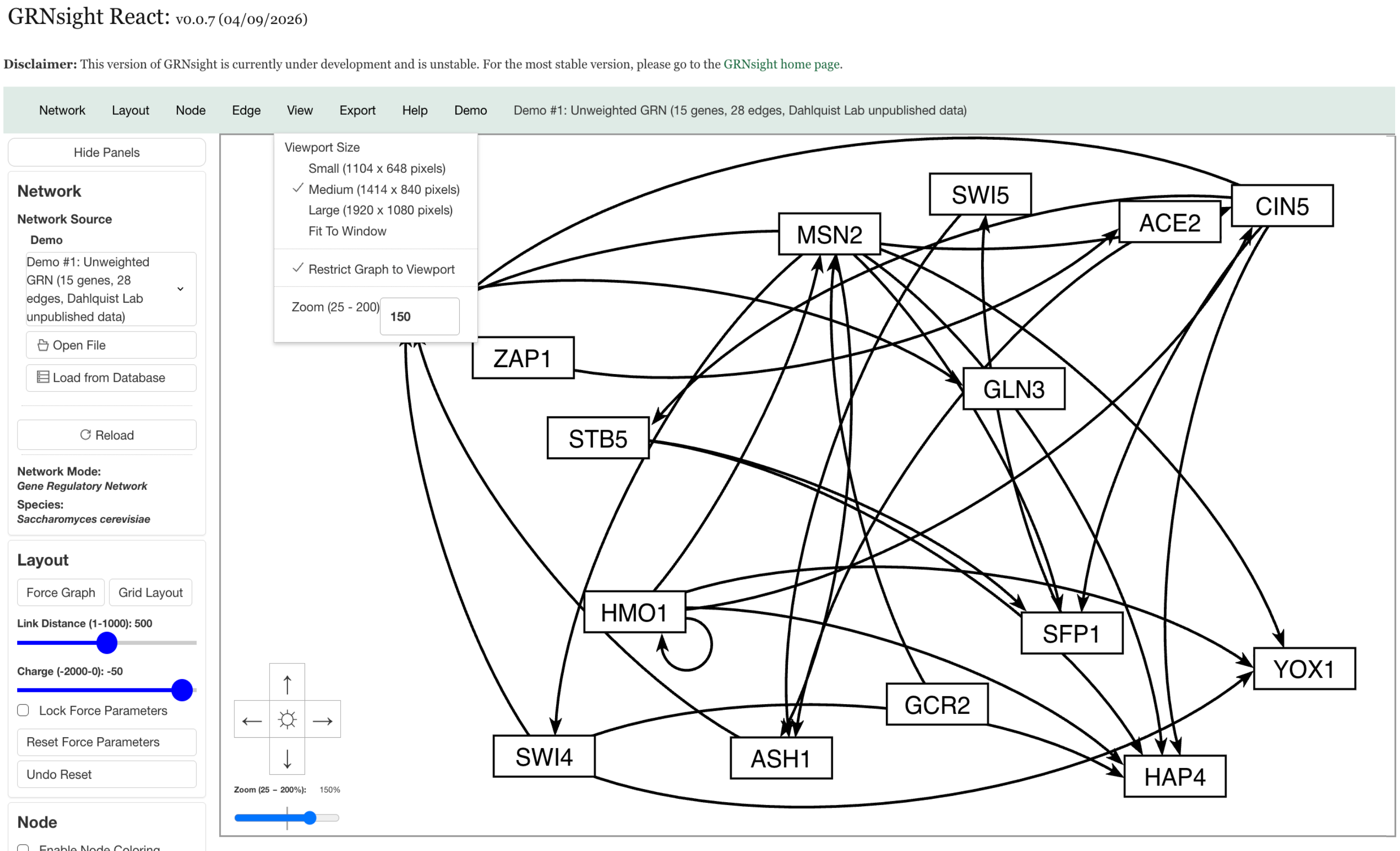
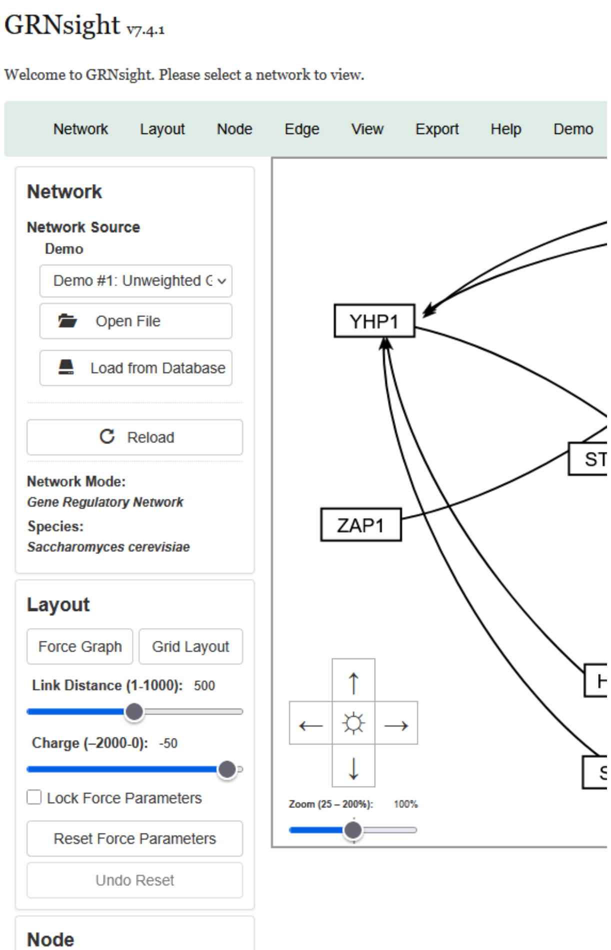
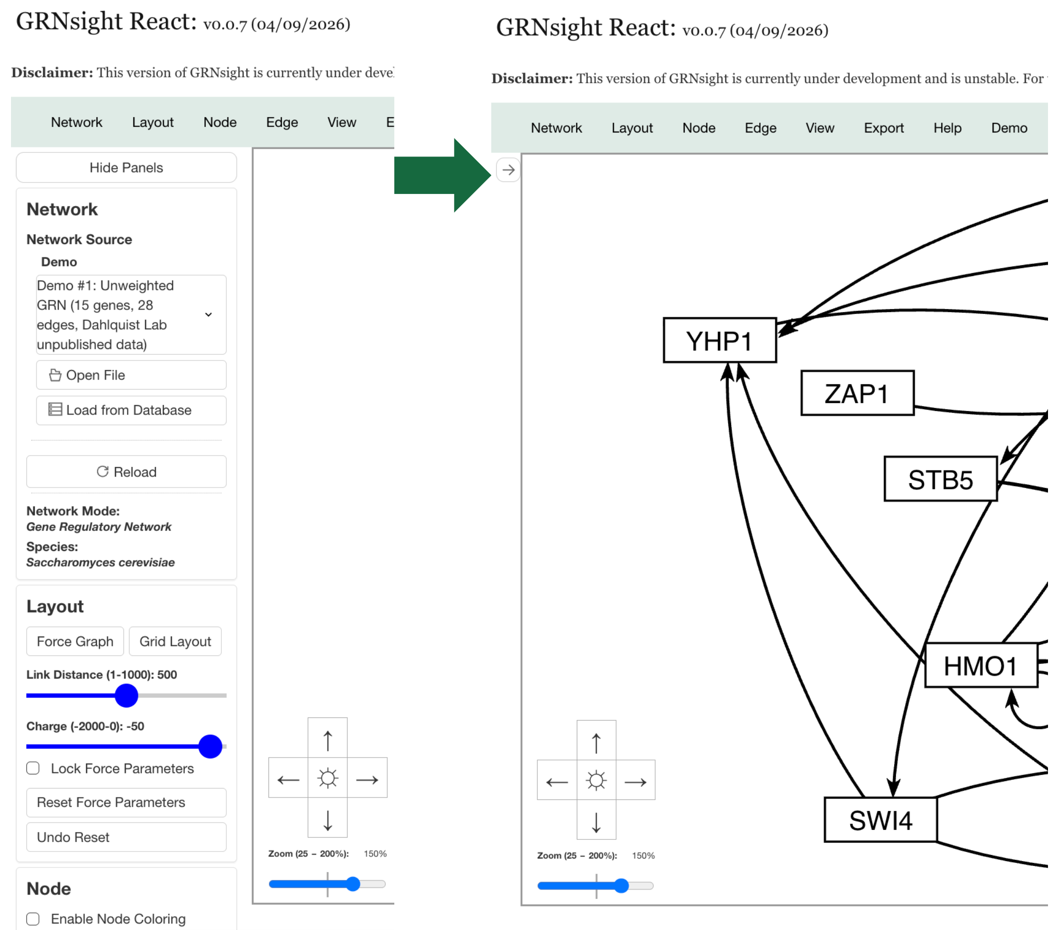
# GRNsight React went live on October 15, 2025, and can now display graphs



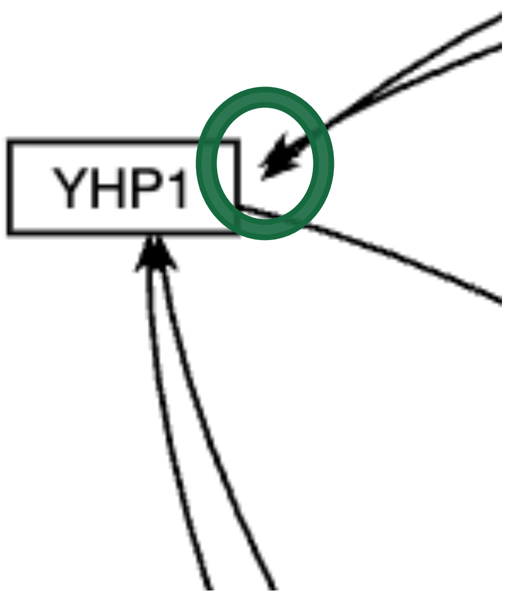

- https://dondi.github.io/GRNsight/react-thesis-4081/
- Viewport size, panning, and zoom are available graph features

# React allows the fast implementation of collapsible side panels

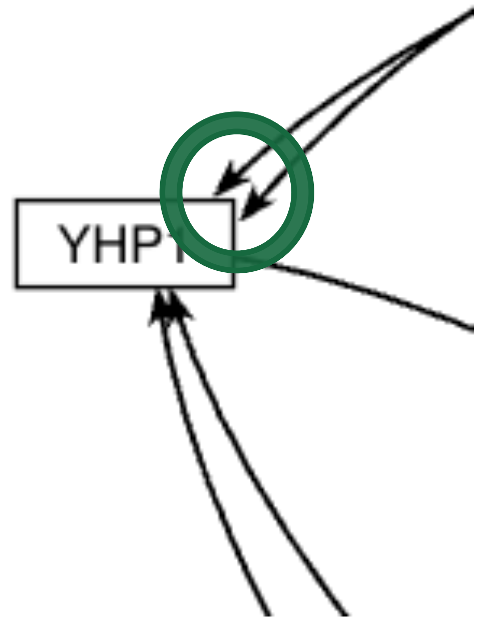


## The high-level code review facilitated the fixing of a long-standing arrowhead bug in GRNsight Classic




Classic Arrowheads with space between the perimeter




React Arrowheads touching the perimeter of the node

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# Acknowledgements

**Pictured Left to Right:**

- Ngoc Kim Ngan Tran
- Milka Y. Zekarias
- Amelie T. Dinh
- Jia S. Garcia
- Cindy L. Tong
- Kam D. Dahlquist
- Alex J. Miller
- John David N. Dionisio

**LMU Department of Biology**

**HHMI STEM Research Learning Community**


http://dondi.github.io/GRNsight/

https://github.com/dondi/GRNsight

## References


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- “PDB101: Molecule of the Month: Cyclin and Cyclin-dependent Kinase.” *RCSB: PDB-101*, pdb101.rcsb.org/motm/236.
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